

22kW/25kW ultra-wide voltage V2X Bidirectional DCDC EV Charger User manual

**Shenzhen UUGreenPower Co., Ltd
2026**

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4	1.03	Revised section description	2026-5-19		

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Safety usage instructions

Before using the product, please carefully read the safety instructions to avoid unnecessary damage and disputes.

- The installation, use, and operating environmental conditions of the product must not exceed the normal operating conditions specified in this manual.
- The product installation should be securely fastened without shaking to prevent potential safety accidents caused by detachment.
- The size of the external cable of the product should be selected within the specified wire diameter range, otherwise it will affect the waterproof reliability of the product.
- Before the product is officially put into use, the wiring of the incoming and outgoing connecting cables should be checked for correctness and firmness without looseness.
- The casing and internal grounding devices of the chassis should be reliably grounded to ensure the continuity and reliability of the protective grounding.
- Before charging and discharging the gun, check and confirm that there is no water accumulation, foreign objects or deformation in the charging plug and vehicle socket.
- It is strictly prohibited to insert or remove the charging gun or open V2X charger door during the charging and discharging process to avoid accidents.
- In case of an emergency during the charging and discharging process, please press the red emergency stop button located on the lower right side.
- Non professionals are strictly prohibited from opening the chassis. Professional personnel engaged in equipment installation or maintenance must have qualifications for high-voltage, direct current, and other operations, strictly comply with relevant industry safety regulations, and operate in accordance with the provisions of this manual and on-site operating procedures.
- Before maintenance, the equipment should be completely stopped to prevent residual electrical energy from causing electric shock, and the corresponding higher-level low-voltage distribution switch should be disconnected.
- The left and right sides of V2X charger should be ventilated for heat dissipation, and it is not allowed to stack debris or flammable materials near them.

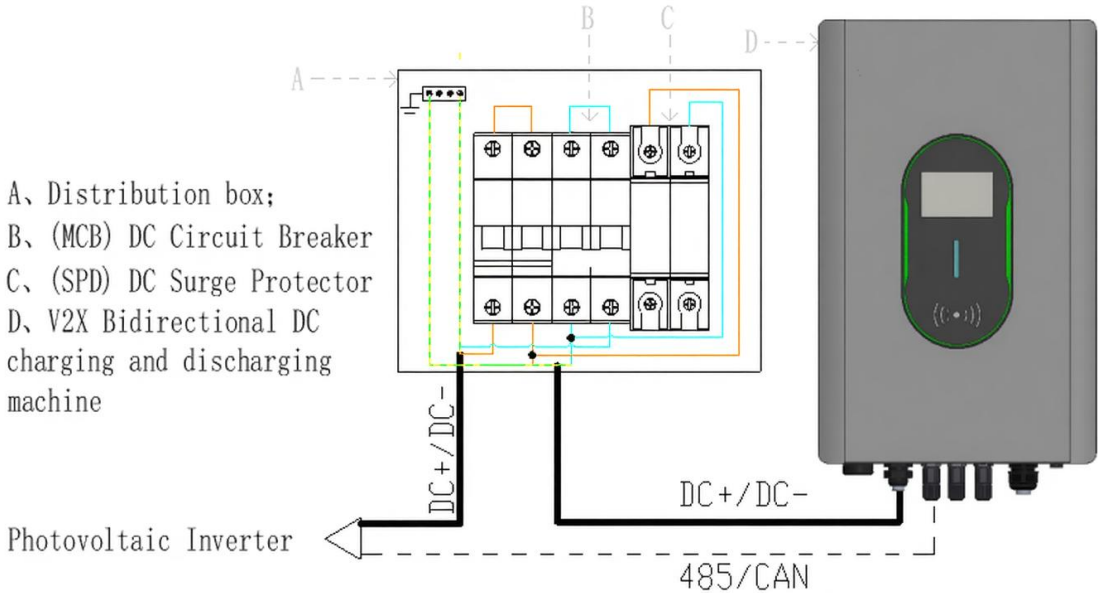
Important Notice

- This product is only allowed to be maintained by trained and qualified electrical personnel. Operators should be familiar with all the precautions and safety warnings in this manual, and equipped with sufficient safety equipment and basic first aid capabilities.
- Our company shall not be held legally responsible for safety accidents or damages caused by violations of the operating manual.
- Replacing internal components of the product without the manufacturer's consent will be considered a serious violation of regulations, and our company will not be held legally responsible for any consequences arising from this behavior.
- Please pay attention to the danger warning signs on the equipment:

 danger	✧ The components of the system carry high voltage during operation, and direct contact will result in personal injury or death. ✧ There are safety hazards during this operation, and failure to follow the instructions may result in fatal danger.
 Warning	✧ There may be hazardous voltage present in the components and connection contacts inside the device! Touching live parts may cause serious bodily injury or even death due to electric shock or burns. ✧ The prompts indicate potential danger during operation. Failure to follow the prompts may result in personal injury or fatal accidents.

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SHENZHEN UUGREENPOWER CO., LTD	5

1 System Overview

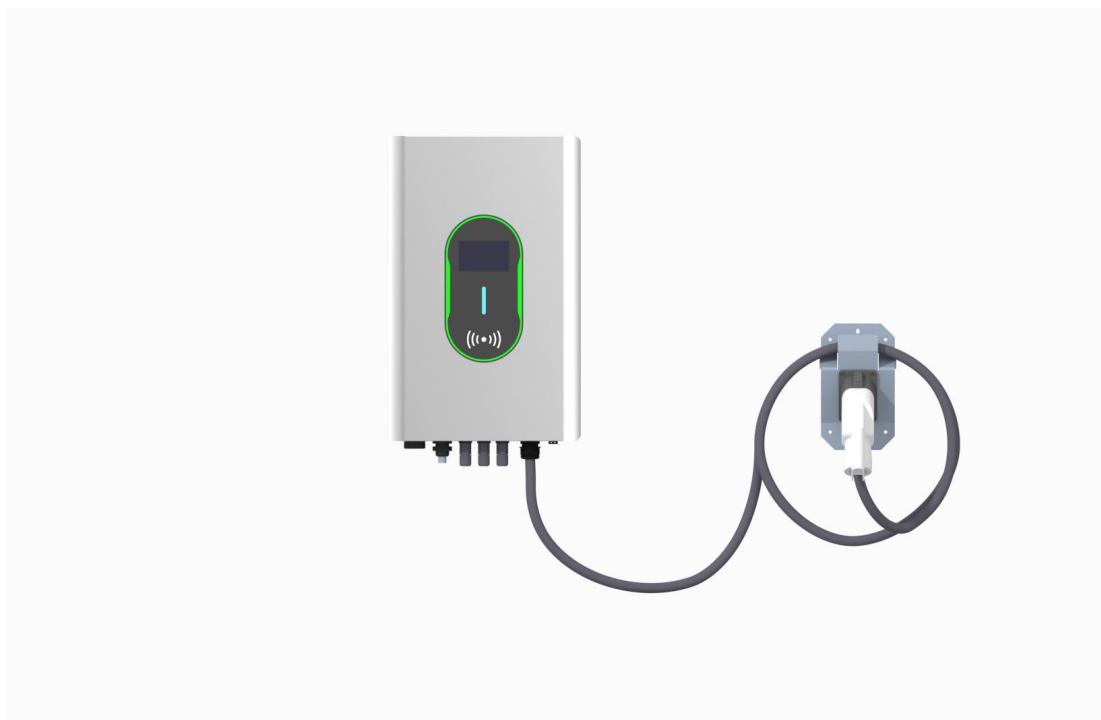
1.1 Product Introduction

The 22kW/25kW V2X Bidirectional DC/DC EV Charger (referred to as the "V2X charger") is a bidirectional DC/DC charging and discharging system that can be DC-coupled with hybrid inverters to charge and discharge the power batteries of electric vehicles for household V2G/V2H applications. It not only meets the daily charging needs of EVs but also functions as an energy storage unit in the vehicle's battery system, participating in grid peak shaving, suppressing load fluctuations, and enabling orderly charging and efficient management of power demand. The V2X charger features a highly integrated IP65-rated design, incorporating power input, power conversion unit, output switch unit, charging control unit, metering unit, charging interface, and human-computer interaction interface. It offers comprehensive functions including display, card reading, emergency stop, metering and billing, fault protection, and data and event record query.

Model difference

The UBC22K1000-AY1-CCS2D, UBC22K1000-AY1-CCS2D-G2 have the same structure, hardware and software except the minor difference as below table

Difference \ Model	UBC22K1000-AY1-CCS2D (N)	UBC22K1000-AY1-CCS2D (N) -G2
	UBC25K1000-AY1-CCS2D (N)	UBC25K1000-AY1-CCS2D (N) -G2
Shunt model	FL2-50	ESFL-2-150
DC meter model	JSY-MK-224	DCM230-2-150



Declaration:

Due to the continuous updating and improvement of series products and technical standards, some content in this material may not fully match the actual latest products. We apologize for any inconvenience caused. To inquire about product updates, please log in to our company website or contact our business representative directly.

1.2 Model Description

The naming specifications for V2X chargers are as follows:

22kW V2X:

UBC22K1000-AY1-CCS2D	22kW V2X, Standard Edition,with touch screen and onboard DC meter
UBC22K1000-AY1-CCS2D-G2	22kW V2X, with touch screen an MID DC meter
UBC22K1000-AY1-CCS2DN	22kW V2X, No screen and onboard DC meter
UBC22K1000-AY1-CCS2DN-G2	22kW V2X, No screen and MID DC meter

25kW V2X:

UBC25K1000-AY1-CCS2D	25kW V2X, Standard Edition,with touch screen and onboard DC meter
UBC25K1000-AY1-CCS2D-G2	25kW V2X, with touch screen and MID DC meter
UBC25K1000-AY1-CCS2DN	25kW V2X, No screen, onboard DC meter
UBC25K1000-AY1-CCS2DN-G2	25kW V2X, No screen, MID DC meter

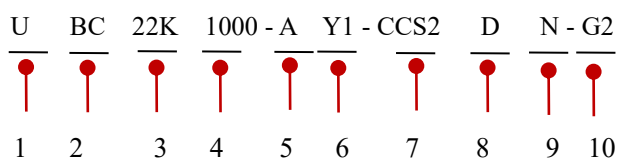


Table 1-1 Model Description

serial number	meaning	explanation
1	Company name	U represents UUGreenPower
2	Product Type	BC represents bidirectional charger
3	Rated power	22K represents 22kW, 25K represents 25kW
4	Rated output voltage	1000 represents 1000VDC
5	Cooling method	A represents independent air duct heat dissipation method (IP65)
6	Product structure form	Y1 represents Integrated charging pile, single gun
7	Charging interface type	CCS2 represents European standard CCS2 charging interface
8	Supplementary marker 1	D represents DC input, DC output
9	Supplementary marker 2	N represents no screen , not written represents standard with touch screen
10	Supplementary marker 3	G2 represents optional MID DC meter, not written represents default onboard DC meter

1.3 Performance Characteristics

- DC coupling with hybrid inverters, enabling high-efficiency utilization of PV- generated green electricity to provide 22kW/25kW DC fast charging for electric vehicles.
- Supports V2H (Vehicle-to-Home) functionality, allowing electric vehicle power batteries to supply power to household loads in reverse, and can be configured for V2G (Vehicle-to-Grid) operation to feed electrical energy back to the grid in conjunction with inverters.
- Supports up to 22kW/25kW bidirectional DC charging and discharging, capable of increasing the vehicle's cruising range by approximately 120 kilometers/150 kilometers per hour.
- Ultra-wide charging and discharging voltage range (EV side: 200-1000VDC), compatible with all CCS European and American standard models.
- Independent air duct heat dissipation design and an IP65-rated enclosure, ensuring maintenance-free operation and an ultra-long service life.
- Multiple protection mechanisms, including high-frequency isolation, insulation detection, surge protection, and leakage current detection, are integrated for enhanced safety.
- Operates reliably within an ultra-wide temperature range of -25°C to 50°C.
- Supports the European standard OCPP1.6J (default)/2.0.1 (Ready) platform protocol and can work with inverters to enable APP-based reservation or charge/discharge scheduling based on electricity prices.
- Support OTA intelligent hardware upgrades, with default support for DIN70121/15118-2 protocols and future software upgrades planned for 15118-20 standard V2G protocol.
- Internal transformer isolation and an output voltage discharge circuit provide high electrical protection.
- With a standby power consumption of less than 15W and a maximum charging/discharging efficiency of 97%, highly energy-efficient.
- The DC BUS side voltage range is 240-980Vdc, and the EV side voltage range is 200-1000Vdc in rectification mode and 300-1000Vdc in inverter mode.
- IP65-rated design ensures safety, stability, and reliability, with an MTBF of $\geq 360,000$ hours.
- Noise levels are kept below 55dB (25 °C, full load rated output)
- Supports DC side voltage regulation, current limiting, and current balancing functions.
- Built-in hardware emergency stop and battery voltage acquisition functions are included.
- supports QR code scanning, card swiping, and OTA remote upgrades, making it suitable for both operational and home scenarios.
- Optional EV side black start function is available to handle extreme conditions such as power outages, continuous rainy days without photovoltaic power generation, and battery depletion, allowing the EV battery to independently provide stable and long-term backup power for important household loads.

1.4 Technical Parameters

Table 1-2 Technical parameters of charging and discharging piles

indicator		22kWV2X	25kWV2X	remark
Charging Mode	Input connection	DC+/DC-/PE		DC input
	Input voltage range	240-980Vdc, 540-980V constant power 22kW, 240-540Vdc Constant 44A	240-980Vdc, 600-980V constant power 25kw ≤ 40 °C, constant 44A below 600V	
	Input current range	0~44A max	0~44A max	
	Output voltage range	200-1000VDC (derating below 300Vdc)	200-1000VDC (derating below 312.5Vdc)	
	Output current range	0~73.3A max	0~80Amax	
	Output power	22kW	25kW	
	Charging efficiency	97% (Peak)		
Discharging Mode	Input voltage range	300-1000Vdc	300-1000Vdc	
	Input current range	0~80A max	0~80Amax	
	Output current range	0~44A max	0~44A max	
	Output voltage range	240-980Vdc, 540-980V constant 22kW, 240-540Vdc Constant 44A	240-980Vdc, 600-980V constant 25kw, below 600V constant 44A	
	Discharge efficiency	97% (Peak)		
Communication	Wireless comm. methods	4G/WiFi/Bluetooth (optional)		
	Local comm. methods	CAN/RS485		
	EV comm. protocol	ISO15118-2, DIN 70121, ISO15118-20 (Forward)		
	Back-end protocol	OCPP1.6j/2.0.1, Modbus-TCP & Modbus-RTU		
	Activation method	RFID/VIN/No authentication		
Mechanical	Dimension	660*420*207 (W*H*D)		
	Weight	47kg		
	Cooling method	Independent air duct, forced air cooling		
	Noise	<55dB (1m)		rated condition rated
	Protection level	IP65		
	Installation	Wall/pedestal mounted		
Function	Operating temperature	-25°C~+50°C	-25°C~+50°C	
	Humidity	≤95%RH no condensation		
	Altitude	2000m		
	Charging interface	CCS1/CCS2 (default 5 meters)		
	Alarm mode	Sound and light alarms		
	Protection	Short-circuit protection/ Over / Under voltage protection/Overload protection/ Over temperature protection/Reverse polarity protection/Welded contactor check		
	User interfaces	5-inch touch screen display (optional) , LED indicator light (customizable), RFID card swiping (authorized charging and discharging), APP (inverter protocol needs to be connected)		
	Upgrade and Diagnosis	Upgrade via USB drive or local host computer Support OTA, Remote diagnostics		

1.5 Function Configuration

Table 1-3 Function Configuration of V2X charger

serial number	project	Technical parameter values
1	Display and input functions	Optional, 5-inch color touch screen
2	Card reading function	Standard configuration, card reader and charging card
3	Indicator light progress display function	Standard configuration, charging and discharging progress indicator light (customizable)
4	incoming circuit breaker	Optional, 50A with leakage protection circuit breaker
5	External incoming cable	Standard configuration of 2 meters, length can be customized
6	Emergency stop function	Standard emergency stop button
7	Remove the electrolytic lock function	Standard configuration, charging gun electronic lock automatically unlocks upon power failure
8	Power down hold function	Standard configuration, save data after power failure
9	Charging method	Standard local card swiping authorization, or combined with Inverter APP authorization to start charging and discharging
10	Pre-charging function	standard configuration
11	Vehicle communication function	Standard configuration, with communication function with vehicle BMS or controller
12	Vehicle plug temperature monitoring function	standard configuration
13	Vehicle plug locking function	standard configuration
14	Insulation monitoring function	standard configuration
15	DC output circuit short circuit detection function	standard configuration
16	DC contactor contact detection function	standard configuration
17	Fault recording function	Standard configuration, saving the time and type of fault occurrence for users to query
18	cooling method	Standard fan, forced air cooling
19	triple protection	Standard configuration, three proof treatment
20	rust protection	Standard configuration, adopting double-layer rust prevention measures
21	windproof protection	Standard configuration, capable of withstanding a maximum wind speed of 28.5m/s
22	Anti-theft protection	Standard configuration, equipped with anti-theft lock and anti-theft screw
23	protection function	Equipped with input side overvoltage, undervoltage, and onboard lightning protection;Over temperature protection;Door opening protection, DC bus anti reverse connection protection;Overvoltage and over temperature protection for individual batteries;Charging overvoltage, overcurrent, and short circuit protection;DC side insulation monitoring and protection;

Table 1-4 V2X charger indicator light language

Charger status	light language
Idle (no gun inserted)	Sky blue (always bright)
Standby (inserting gun)	Sky blue (always bright)
Charging	Green (always on)
Discharging	Blue (constantly bright)
Charging/discharging completed	Light green (constantly bright)
fault	Red (always on)
alert	Yellow (constantly bright)
Charger locking	Coral color (constantly bright)
Charger not connected to the internet	Purple (constantly bright)
Upgrade DC	Orange (progress bar)
Upgrade PFC	Orange red (progress bar)

1.6 Operating Instructions

1.Charging Connection

Remove the charging gun plug and connect it to the vehicle's charging socket.

2.Select Charging/Discharging Mode

- If a touch screen is available, tap or swipe the card on the screen to initiate charging/discharging.
- If no touch screen is present, the inverter can send a start command, and you can operate the start function via the inverter's APP.

3.Start Charging/Discharging

Begin the process and monitor the charging/discharging status in real-time through the touch screen (if available) or the inverter APP.

4.Stop Charging/Discharging

Tap the touch screen (if available), use the inverter APP, or swipe the card again to end the charging/discharging session.

5.Complete the Process

Unplug the charging gun plug and return it to the charging/discharging station socket to finalize the process.

Emergency Handling

If an emergency occurs during charging/discharging (e.g., abnormal noise, smoke, fire), immediately press the emergency stop button, disconnect the power supply, and contact on-site management personnel.

2 Composition of Charger

2.1 System Composition

The V2X charging system primarily comprises a DC input, a power conversion unit (bidirectional DC/DC converter), an output switch unit, a charging cable, and a vehicle plug. It also includes a control power supply, a charging control unit, a human-machine interaction unit, and a metering unit. The charging control unit centrally manages all system components, and data from each unit is transmitted to the charging control unit via RS485 communication, CAN communication, digital I/O signals, and analog-to-digital (AD) sampling.

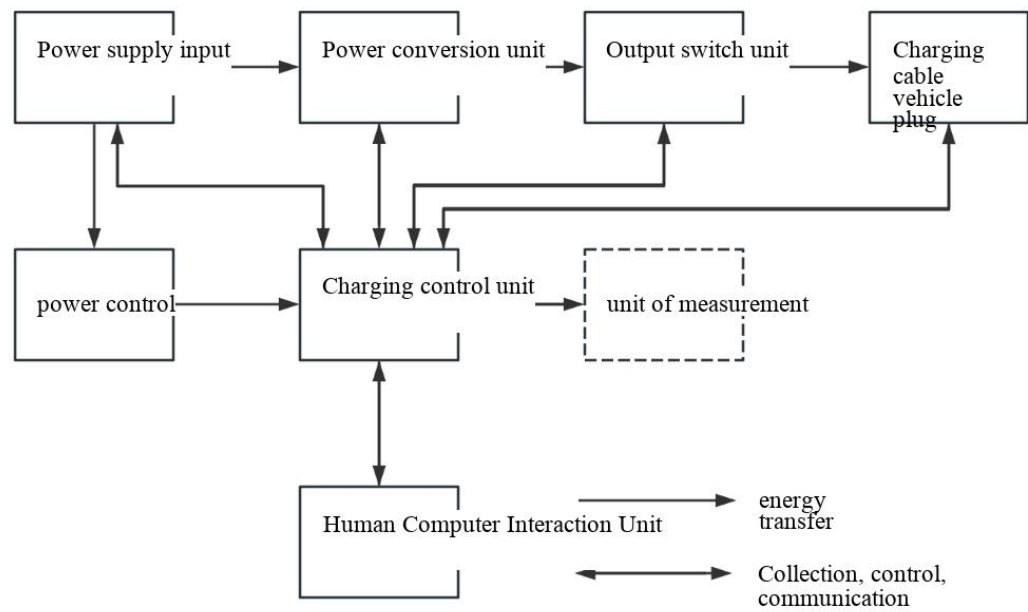


Figure 2-1 V2X Charger: Composition, Principle, and Block Diagram

2.2 Equipment Configuration

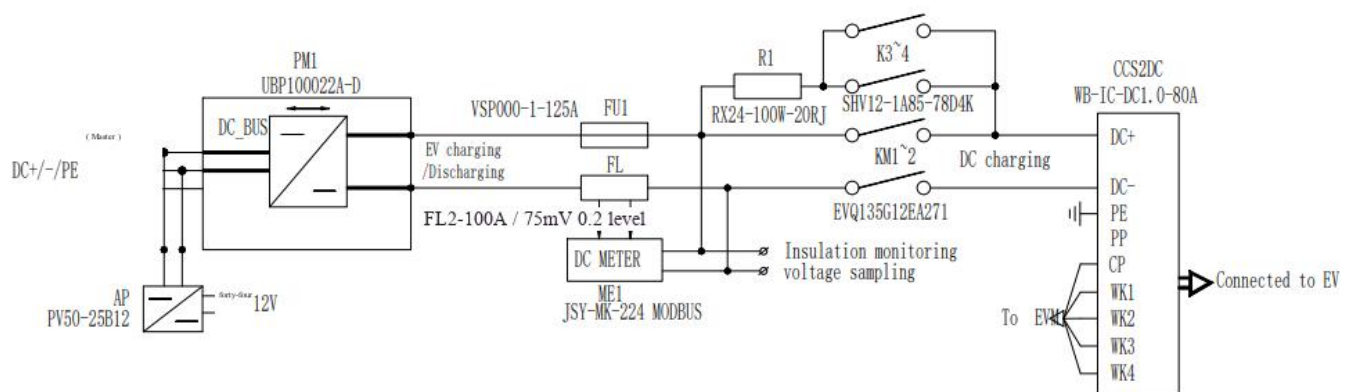
Table 2-1 V2X Charger Configuration

serial number	building block	Device Configuration
1	Power supply input	DC circuit breaker (optional), lightning arrester (optional)
2	Power conversion unit	22kW/25kW isolated bidirectional DCDC module
3	Charging cable and vehicle plug	European standard CCS2 DC charging gun cable, default 5 meters, length can be customized
4	power control	12V Switching Power Supply
5	Charging control unit	CCS02 Charge and Discharge Charger Control Unit
6	communication module	Standard WIFI and Bluetooth 2-in-1 (enabled by default with APP) Optional 4G module (built-in in the system cabinet, with a SIM card slot for external use)
7	Output switch unit	High voltage DC contactor * 2
		High-voltage fuse

serial number	building block	Device Configuration
8	Human Computer Interaction Unit	Status indicator light (standard), customizable light language
		5-inch touch screen (standard), customizable charging page language switching, startup animation, logo, etc Optional no screen version, only retaining indicator lights
9	card reader	Swipe card reader (customizable with logo)
10	Offline card	Offline IC card (card swiping authorization for charging and discharging, or card free activation)
111	unit of measurement	(Standard) 1.0 level onboard DC meter (Optional) 1.0 level MID DC meter, built-in in the system cabinet
12	Emergency protection system	Metal Self-Locking Emergency Stop Button
13	DC Arc Detection (Optional)	AFCI DC arc detection, built-in in the system cabinet
14	DC Circuit Breaker (Optional)	Specification 1000V/50A/4P, connected between V2X and inverter, can be externally installed
15	Black Start Component (Optional)	Built into the system cabinet, with a reserved black start button BLACK START for external use

2.3 Electrical Principles

The output configuration of V2X charger includes an 80A European standard CCS2 DC charging gun. A DC energy meter is installed in the DC power supply circuit for independent metering and billing. A high-voltage DC contactor is designed on the DC output side to control the closing and opening of the charging circuit, as well as to forcibly disconnect the charging circuit in emergency situations, ensuring personal safety.



2.4 Appearance Size

2.4.1Physical Appearance



Figure 2-2 Physical Image 1 of V2X Charger (With Screen Version)



Figure 2-3 Physical Image 2 of V2X Charger (No Screen Version)

2.4.2Product Dimension

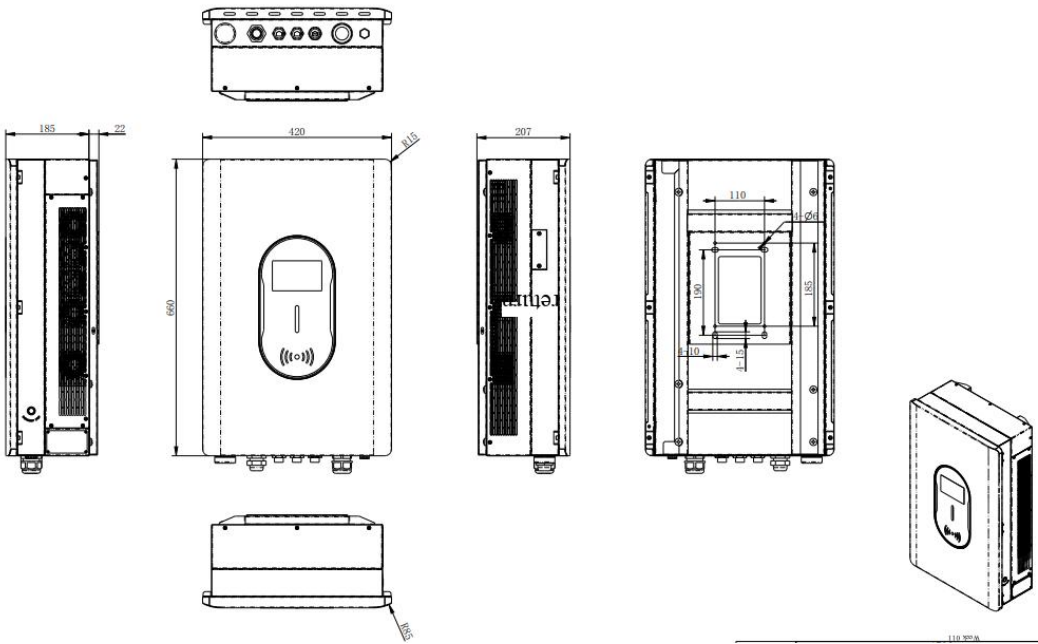


Figure 2-4 Dimension diagram of V2X charger (with screen version)

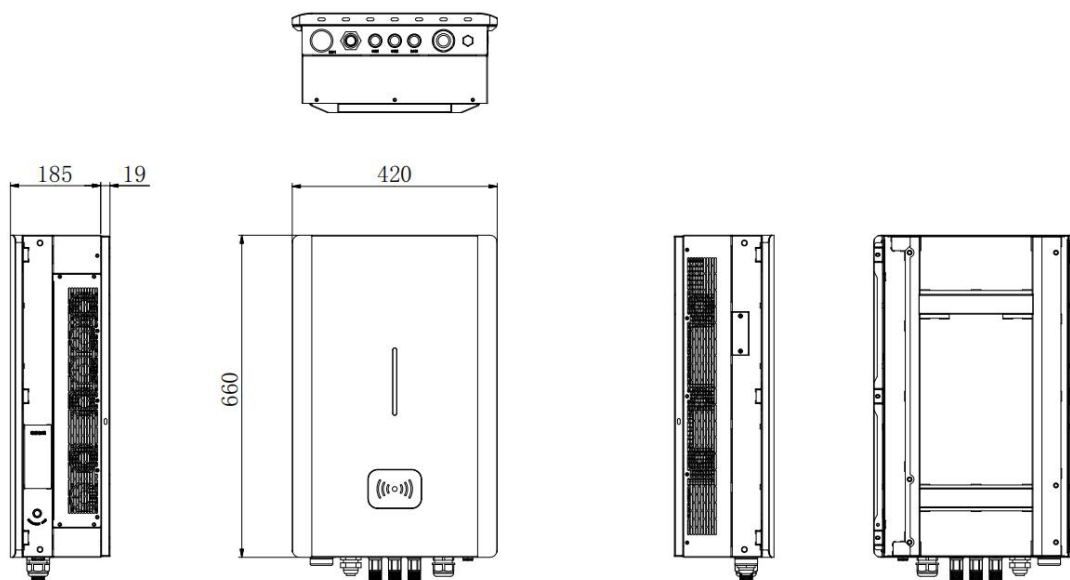


Figure 2-5 Dimension diagram of V2X charger (No screen version)

2.4.3 Interface description Port description

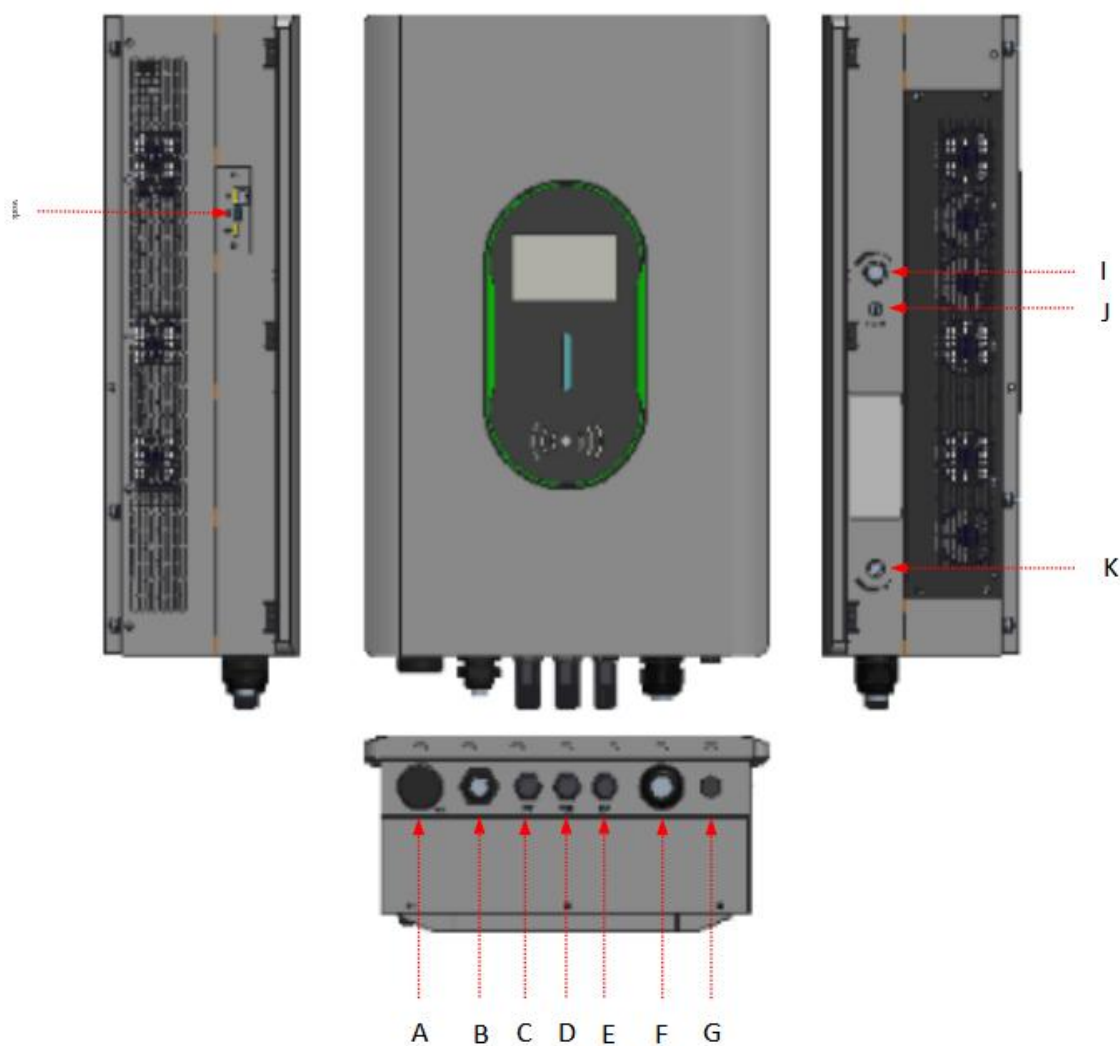


Figure 2-6 Schematic diagram of V2X charger interface

Object	silk-screen	function	PIN
A	WIFI/4G antenna	WIFI/4G antenna, built-in WIFI/4G module comes standard with WIFI and Bluetooth (two in one), 4G optional	/
B	DC incoming line	Connect the inverter DC bus input port, default is 2 meters	3*10mm ²
C	COM1	Used for communication with inverters based on Modbus RTU protocol (RS485)	485A:PIN 4 485B:PIN 5 CANH:PIN 3 CANL:PIN 6 DSP-485A:PIN 1 DSP-485B:PIN 2 DSP-IN:PIN 7 DSP-GND:PIN 8
D	COM2	Used for reservation or merging	485A:PIN 4 485B:PIN 5 CANH:PIN 3 CANL:PIN 6 DSP-485A:PIN 1 DSP-485B:PIN 2 DSP-IN:PIN 7 DSP-GND:PIN 8
E	RJ45	RJ45 wired network port	/
F	EV gun line	CCS2 DC Charging Cable	/
G	/	Vent valve	/
H	UI	RJ45 Ethernet port/capture message for retrieving charging logs	Ethernet port
	APP	USB interface/program upgrade upgrades	USB port
	SIM	4G card socket/network connection	Insert 4G card
I	BLACK START	black start button	/
J	Type C	Black start external power bank power supply interface	/
K	Emergency stop button	Emergency stop	/

Attention: Interface H needs to loosen the screw to open the protective cover
Port H loosen the screw to open the protective cover

2.4.4 Packing Materials List

serial number	Material Name	Specification Model	unit	quantity
1	cabinet	System Unit	Pcs	1
2	Offline card	Offline IC card	Pcs	1
3	plastic part	European standard DC gun holder	Pcs	1
5	Sheet metal parts	Sheet metal parts, 0003 Euro Spear Seat Bracket, 254.6 * 150 * 62	Pcs	1
6	Sheet metal parts	Sheet metal parts, 0136 wall mounted fixed plate, 250 * 225 * 22mm	Pcs	1
7	screw	Screw, M10 * 80 expansion screw	Pcs	4
8	screw	Screw, M6 * 50 expansion screw	Pcs	4
8	combination screw	Combination screw, M4 * 12 cross hexagon head double pad combination screw	Pcs	2

2.5 Installation Instructions

2.5.1 Installation precautions

Before installation, it should be confirmed that the environmental conditions and cable specifications meet the following usage requirements for charging and discharging piles:

1. Reserve no less than 600mm of space in front of the V2X charger for maintenance and charging operations.
2. Install fixing screws and ensure ventilation heat dissipation on both sides of the charging and discharging pile, with no obstacles or flammable materials on both sides.
3. When using 4G wireless communication, ensure good 4G signal in the area.
4. The DC incoming cable joint supports a cable outer diameter specification of 17-25mm. Pay attention when selecting the DC incoming cable.
5. The charging gun inlet cable connector supports a cable outer diameter specification of 22-30mm. Pay attention when selecting the gun cable.
6. 4G wireless communication comes standard with a mushroom head antenna or wired communication via Ethernet cable.
7. The lowest point for storing the charging gun head should be between 0.5 meters and 1.5 meters above the ground.
8. Do not allow the use of vehicle charging adapters.
9. The use of cable extension kits is not allowed.

2.5.2 Installation Process

1. Protective grounding wire must be installed first
2. Before operating the equipment, ensure that it is properly grounded
3. The grounding impedance of the equipment should comply with local electrical standards
4. The device must be permanently connected to the protective grounding. Before operation, check the electrical connections to ensure reliable grounding (resistance not exceeding 4 ohms)
5. Install the fixed hanging board on the wall or bracket, using M10 × 80 expansion screws, quantity 4 (provided by the manufacturer)
6. Connect the DC incoming cable to the waterproof distribution box (optional, customer configured) dedicated to charging and discharging piles, or directly connect it to the DC BUS port of the inverter and take waterproof measures
7. If 4G communication is used, insert the SIM card into the card slot of the charging and discharging station monitoring unit
8. After installation, check that the cables are securely fastened and connected.

Structural installation dimensions:

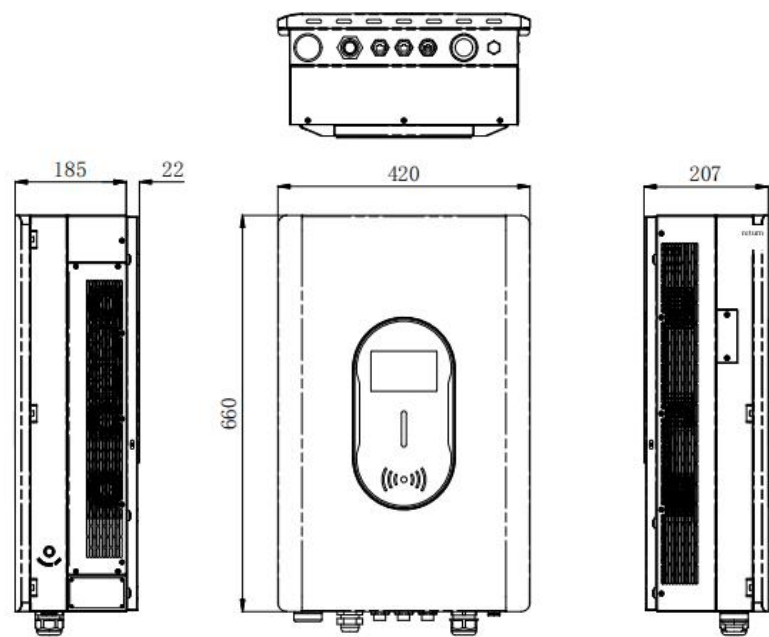


Figure 2-7 Structural Outline Dimensional Diagram

Wall mounted fixed plate installation hole size:

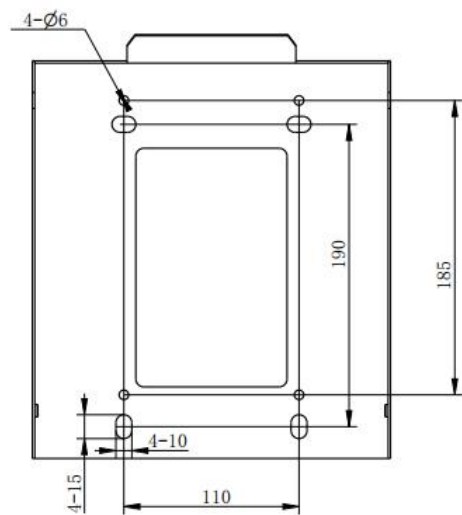


Figure 2-8 Dimension diagram of installation holes for wall mounted fixed plates

Dimensions of gun holder bracket:

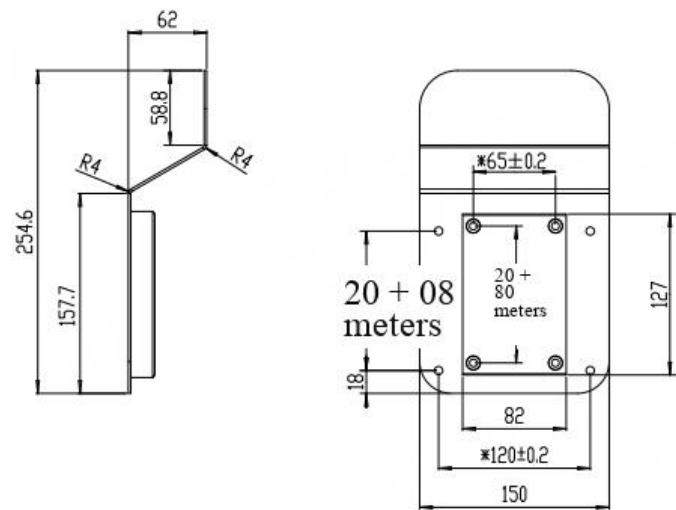


Figure 2-9 Dimension diagram of gun seat bracket

2.5.3 Wall mounted installation steps:

Step 1: 6 M10 * 80 expansion screws are used to fix the wall hanging plate on the wall during wall hanging installation. For the wall with insulation layer, the expansion screws should be extended according to the thickness of the insulation layer. foam brick wall should be equipped with special expansion screws (expansion screws should be provided by the customer).

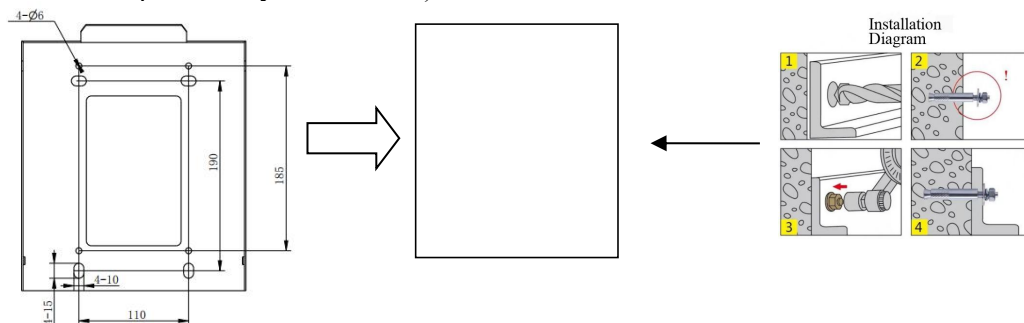


Figure 2-10 Wall mounted fixed plate installed on the wall

Step 2: Fix the gun holder to the wall, use 4 M6 * 50 expansion screws to fix the gun holder bracket to the wall, and the bottom edge should be one meter above the ground.

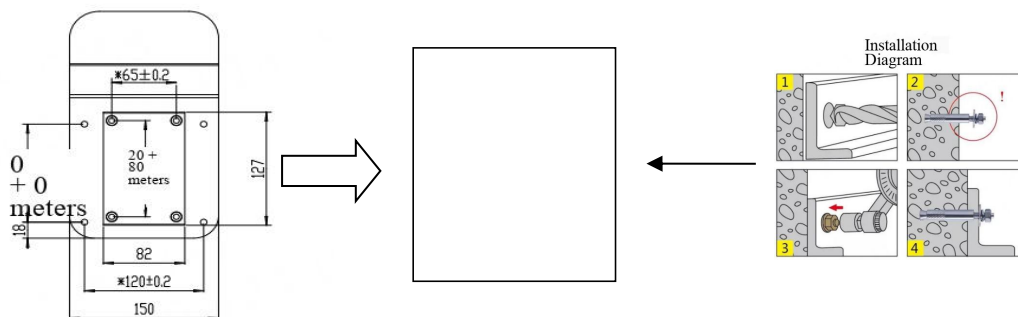


Figure 2-11 Gun holder bracket installed on the wall

Step 3: The bottom edge of the screen should be at least 1.3m (meters) away from the ground. The charging station should be fastened to the wall fixing plate through the back panel, and finally fixed to the wall fixing plate with 2 M4 * 12 three combination screws.

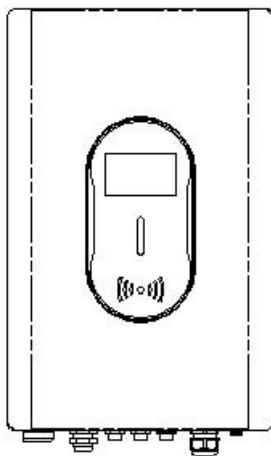


Figure 2-12 Charging and discharging piles fixed on the wall

Step 4: Finally, connect the communication ports connection with hybrid inverter.

2.5.4electrical connection

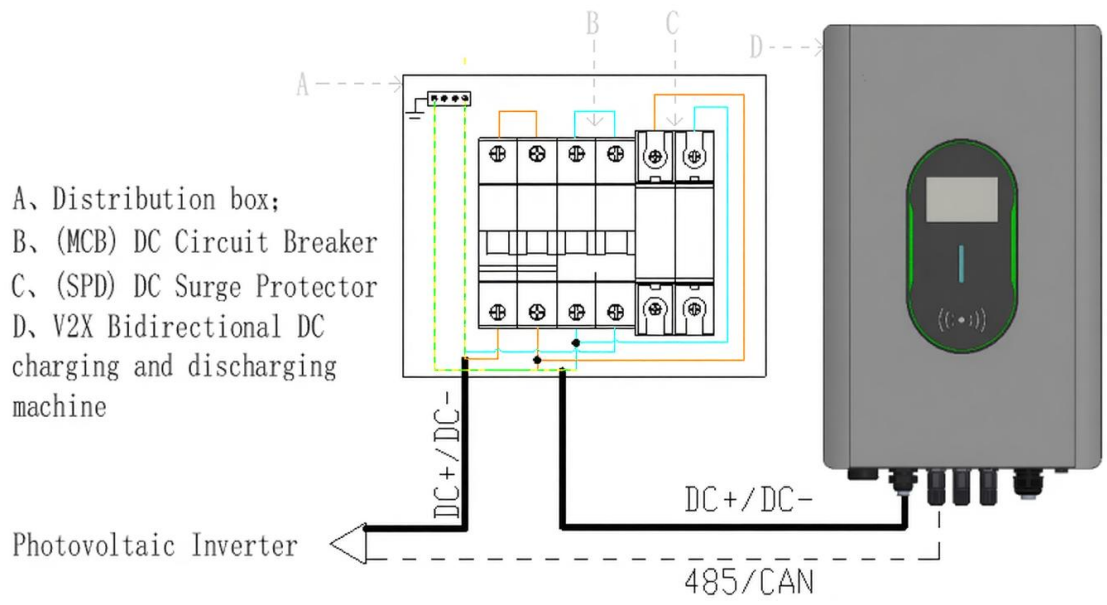


Figure 2-13 Schematic diagram of V2X charger interface

DC input port: Connect the DC+and DC - terminals of the V2X standard input DC cable to the DC+and DC - ports of the distribution box circuit breaker or directly to the DC BUS port of the inverter-

Distribution box, (MCB) DC circuit breaker, can be configured according to actual conditions. When an SPD is installed on the grid side of the Hybrid inverter, the V2X bidirectional DCDC EV charger may not install a (SPD) DC surge protector. When an SPD is not installed on the grid side of the Hybrid inverter, the V2X bidirectional DCDC EV charger needs to install a (SPD) DC surge protector.

(MCB) Direct Current Circuit Breaker Parameters

Rated Voltage (V)	Rated current (A)	Trip Type	Model (Recommended)
DC1000V	50A	C	NDB2Z-63 C50/4P

(SPD) Direct Current Surge Protector Parameters

Rated Voltage (V)	Maximum discharge current (kA)	Rated discharge current (kA)	Model (Recommended)
DC1000V	40kA	20kA	TRS3-C40

EV charging gun cable port: Supports CCS2/CCS1 DC charging gun interface

2.5.5 Communication connection with inverter

V2X chargers support communication with inverters based on Modbus RTU protocol (RS485) or Modbus TCP protocol (wired Ethernet/WIFI).

1. Applicable to all third-party systems or devices that communicate with this device based on Modbus RTU protocol (RS485), refer to the communication diagram below

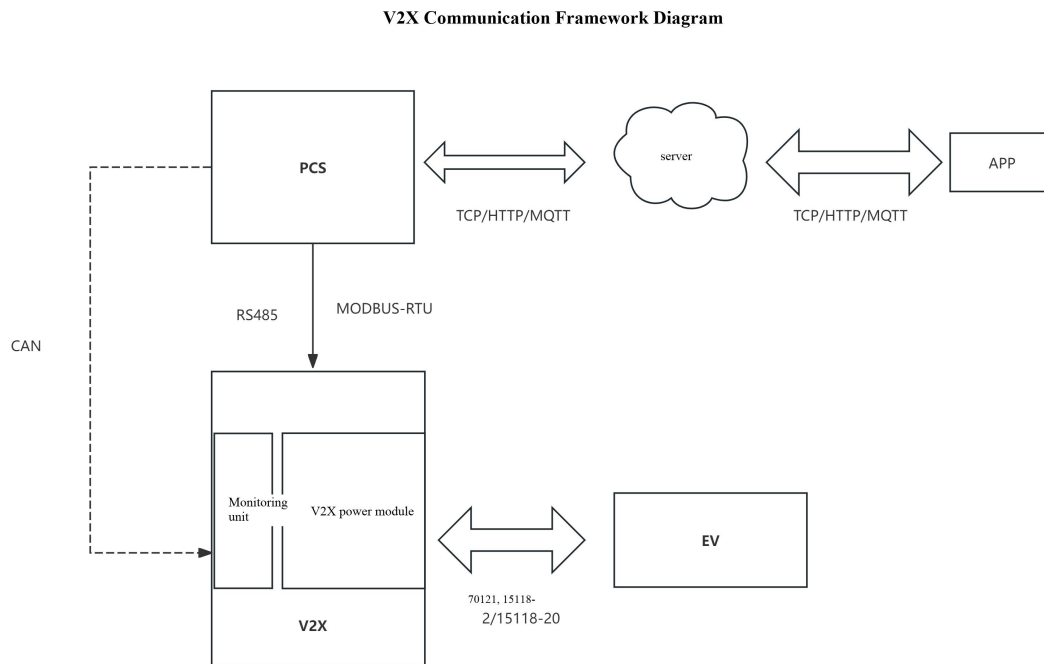


Figure 2-14 Schematic diagram of Modbus RTU communication scheme

2. Applicable to all third-party systems or devices that communicate with this device based on Modbus TCP protocol (wired network port/WIFI), refer to the communication diagram below

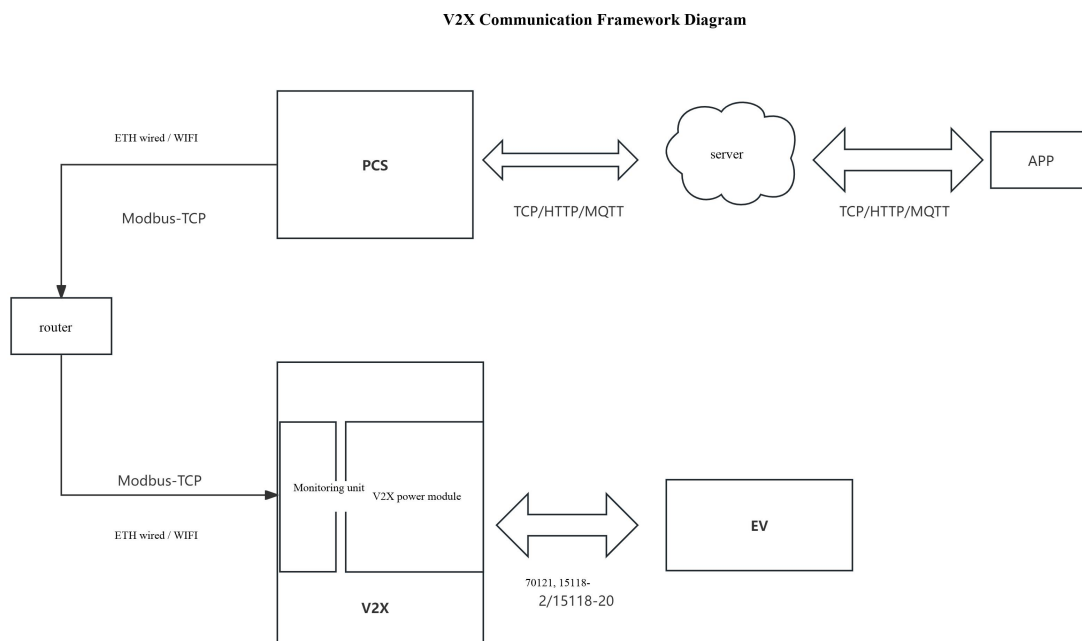


Figure 2-15 Schematic diagram of Modbus TCP communication scheme

Please refer to the attachment for the specific communication protocol

3 System operation interface (optional touch screen)

3.1 System setup instructions

Display specifications

This solution features a 5-inch display with 65K colors and an 800×480 resolution touch screen. The screen delivers clear, complete characters and high contrast, ensuring visibility without reliance on ambient light.

Main Interface

After the monitoring unit is powered on, the system initialization is completed and it enters the main interface of the charging system, which supports both European standard CCS2 DC and American standard CCS1 DC charging protocols. If the internet connection is successful, a QR code will be displayed on the homepage.



If an abnormal alarm condition occurs while the system is in standby mode, the status light of the V2X charger will turn red and an alarm interface will pop up. The red light will indicate the corresponding alarm content, which is convenient for engineers to troubleshoot. The interface is shown in Figure 3-2:

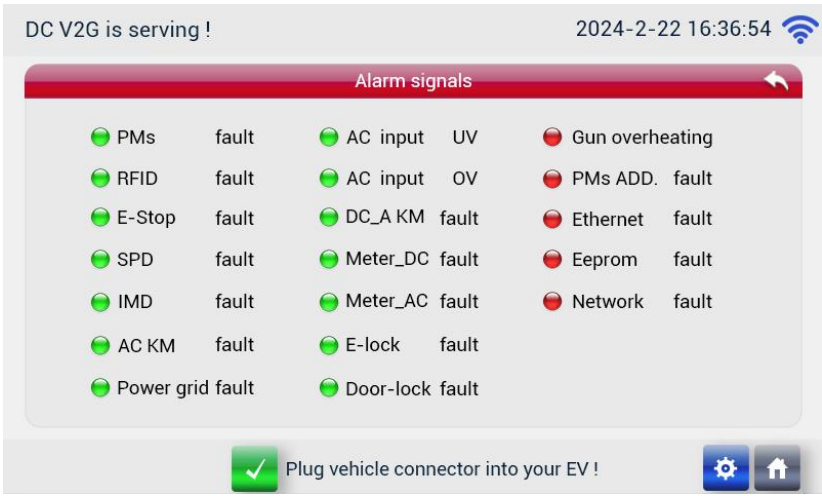


Figure 3-2 Alarm Interface

System Settings Menu

Click the "Settings" button on the homepage of the interface to enter the password input interface. Enter the administrator password to enter the system management page, as shown in Figure 3-3. (Note: The default password is 123456. To exit the password input interface, you must first click the ESC or OK button on the keyboard to make the input cursor disappear, and then click the back button to return to the homepage. If you enter the wrong password, please click on the white display box to make the input cursor appear, and then re-enter.)

Please note that the system parameters should be set according to the actual hardware module!

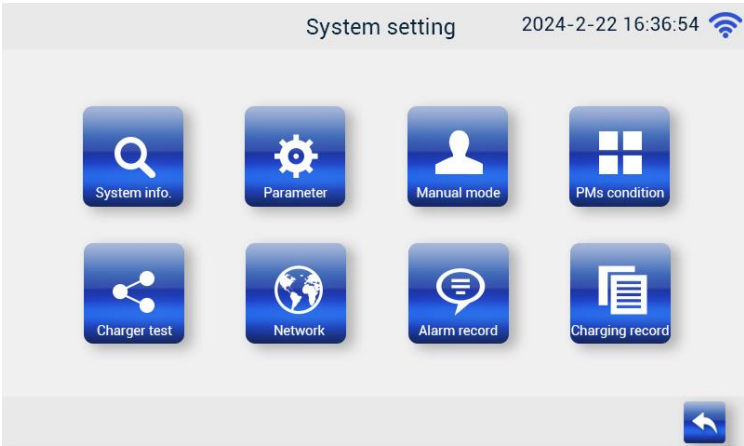


Figure 3-3 System Settings Page

Basic settings and queries

Click the Basic Settings and Query button to set basic parameters such as system time and administrator password, and to query the version numbers of the UI, software, and modules. The interface is shown in Figure 3-4 below:

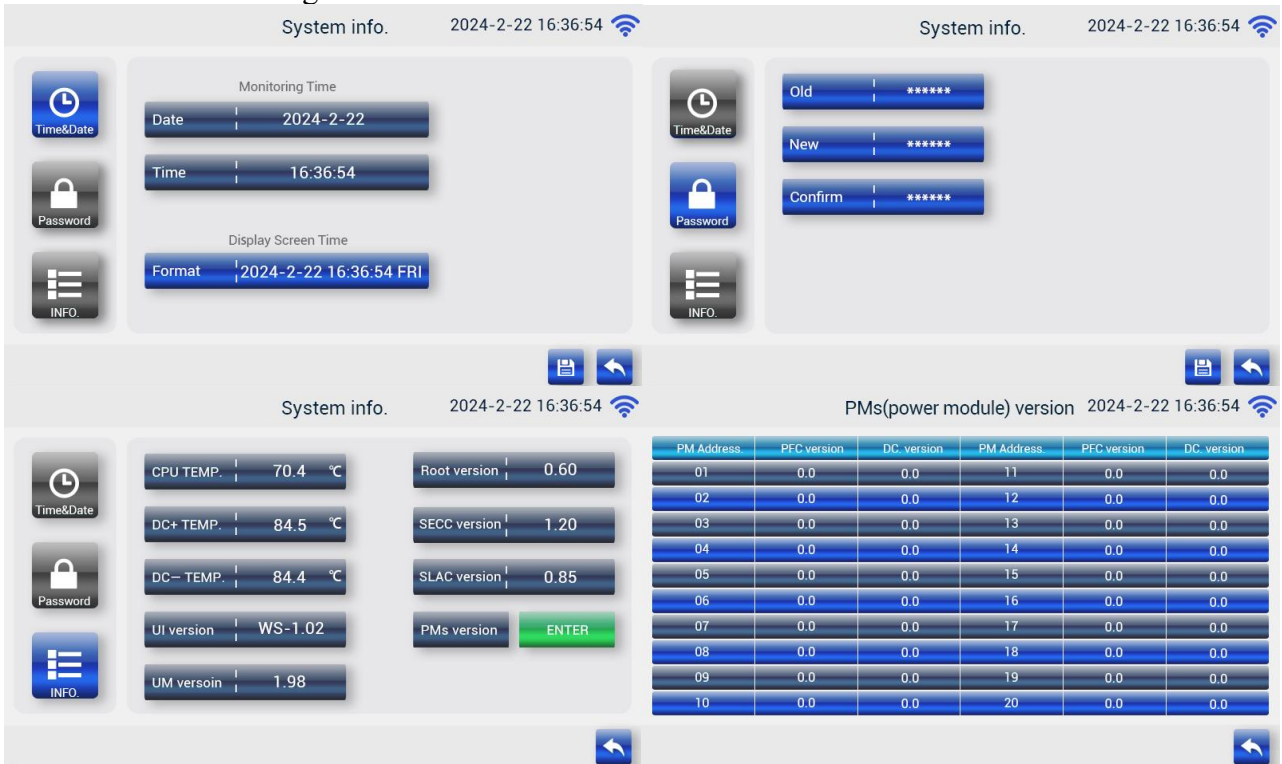


Figure 3-4 Basic Settings and Query Page

- **Monitoring time setting:** Enter a six-digit monitoring date, such as 160101. If the modification is successful, the displayed date will be January 1, 2016. If the monitoring time is entered as a six-digit number, such as 120101, the modified display time will be 12:01:01.
- **Display screen time setting:** When clicking on the display screen time input, the system has fixed the input method as 20??-??-?? ??:???. Customers must enter the correct time in one go before clicking OK, otherwise click ESC.
- **Administrator password setting:** Enter the current password, new password, and confirmed password separately. The password must be a six-digit number. After entering it correctly, click

save and the password will be successfully changed. (The default administrator password is 123456)

- **Information query:** CPU temperature, gun temperature, UI version number, software version number, kernel version number, SECC version number, SLAC version number, and charging power module version number can be queried.

Parameter Settings

Click the parameter settings button to set parameters such as modules, alarm settings, V2X chargers, etc.The interface is shown in Figure 3-5 below:



Figure 3-5 Parameter Settings

① Module settings:

- **Number of modules:** According to the actual situation of the electric pile, set the corresponding number of modules. The quantity of this product is 1;
- **Module type:** Select according to the power module used. This product is 3 and cannot be changed after leaving the factory;
- **Rated power:** Select the corresponding module power level according to the module type. This product is 7:22KW, set to 7, and cannot be changed after leaving the factory;
- **Module communication baud rate:** 0 represents 125Kps, 1 represents 250Kps, 2 represents 500Kps. This product selects 0 and cannot be changed after leaving the factory;
- **Rated current:** It should be set according to the selected module type. The rated current range for a 1000V-30kW module is 0~100A; the rated current range for an 850V-30kW module is 0~60A; the rated current range for a 1000V-7kW module is 0~28A; the rated current range for a 750V-7kW module is 0~23A; the rated current range for an 11kW module is 0~44A; and the rated current range for a 22kW module is 0~80A. This value will limit the maximum output current of a single module;
- **Output power limitation:** The maximum output power during charging can be limited by inputting any non negative integer as required. Setting it to 0 means no output power limitation, and the input value cannot exceed the rated power of the module used;

② Alarm settings:

- **Input overvoltage value:** Set the input overvoltage protection value. If the input voltage exceeds this value, a fault will be reported and charging/discharging will be stopped. The default input voltage overvoltage value is 980V, which can be changed according to the input situation;
- **Input undervoltage value:** Set the input undervoltage protection value. If the input voltage value is lower than this value, a fault will be reported and charging/discharging will be stopped. The default setting for phase voltage undervoltage value is 200V;
- **Output overvoltage value:** Set the output overvoltage protection value. If the output voltage exceeds this value, a fault will be reported and charging/discharging will be stopped. The default setting is 1010V, which can be adjusted according to actual usage requirements;
- **Output undervoltage value:** Set the output undervoltage protection value. If the output voltage is lower than this value, a fault will be reported and charging/discharging will be stopped. The default setting is 150V, and it is not recommended to change it too much;
- **Output overcurrent value:** The default setting is 100A, which can be set according to the power level of the V2X charger, while also considering the maximum current that the charging gun can actually withstand;

③ Electric Charger setting:

- **Electronic lock control mode:** 0 represents general type, 1 represents magnetic holding type;(Special note: The magnetic **holding charging gun must not be set to 0**, otherwise it will burn out the electronic lock of the charging gun. It is recommended to add a 2-3A fuse at the control end of the electronic lock for protection.)
- **DC billing mode:** 1 represents DC meter billing, currently only DC meter billing is available;
- **Time zone setting:** Negative numbers indicate the West time zone, positive numbers indicate the East time zone;For example, input 8 is the East Eighth District, input -8 is the West Eighth District;
- **Seasonal setting:** 0 is winter time, 1 is summer time;
- **Gun temperature sensor types:** divided into none, temperature control switch, and thermistor, which can be selected according to the actual temperature sensor type of the charging gun. After selection, the charging gun temperature can be viewed on the information query interface;
- **DC Charger energy meter protocol types:** 1 represents DLT645, 2 represents Modbus, 3 represents DTL698, users can choose according to different needs;After modifying the protocol type of the energy meter and saving it, the monitoring board needs to be restarted
- **DC energy meter type:** to be compatible with different types of energy meters, select the corresponding energy meter type according to the DC Charger charging billing mode, and also prohibit the use of DC energy meters;(After modifying the type of energy meter and saving it, it is necessary to restart the V2X charging and discharging station)
- **Quantity of colored light beads:** Customers need to input the corresponding quantity of colored light beads based on the Charger structure, with an initial value of 0 and a maximum input of 200;
- **Color light voltage type:** Customers need to choose the correct voltage type based on the

color light type of the pile, with two options: 5V and 12V. They can choose the light language mode or the non light language mode;

- **Color light controller:** 0 represents GPIO port control, 1 represents adding CVT color light controller, customers can choose according to the Charger structure;

Note: Example of parameter setting for DC Charger electric meter. For example, I have a specific model of DCM230 electric meter, which is DC billing and has a communication protocol type of Modbus. When selecting parameters: DC Charger charging billing mode is 1, DC Charger electric energy meter protocol type is 1, and DC Charger electric energy meter type is 4;

④ Equipment Enable Settings:

- **DC contactor status enable:** 0 disable represents that the DC contactor status is not working, 1 enable represents that the DC contactor status is working; Can determine whether the DC contactor is intact;
- **Access control switch enable:** 0 indicates that the function is not working, 1 indicates that it is normally open. When the detection point signal is 0V, there will be a buzzer alarm. 2 indicates that it is normally closed. When the detection signal is 12V, there will be a buzzer alarm;
- **Print SECC details log:** This function is only used for engineer debugging;
- VIN code enable;
- **OCTT testing:** This feature is only used to enable when using the OCTT testing tool;
- Mode switch;

⑤ Electricity price setting:

V2X charging and discharging stations can set peak, peak, normal, and valley electricity prices, as well as service fees, according to different time periods when not connected to the internet; The power loss rate refers to the additional amount of electricity lost during charging, which is borne by the user. The power loss rate is set according to the actual situation of the customer;

⑥ Card settings:

- **Card reader type:** 0 represents suitable for offline cards; 3 represents the prohibition of card reader function; 4 represents online card management when connected to the internet;
- **M1 card sector number and block number settings:** The sector number range is 0-15 and the block number is 0-3, which should be consistent with the selected sector number and block number during recharge. The default sector number is 5 and the block number is 0. This setting is applicable to offline cards;
- **M1 card Key A setting:** The initial password for M1 card Key A is FFFFFFFFFF. If the password is changed using card writing software, enter the corresponding change password in this input box (the default password given by the monitoring unit is ABCDEFFEDCBA). Otherwise, card swiping will fail. This setting is applicable to offline cards;

Note: After the parameter settings are completed, it is necessary to restart the V2X charger to ensure that the various parameters of the monitoring board comply with the designed scheme;

manual control

In the manual control interface, users can activate the manual control module for current transmission without the need for BMS communication, as shown in Figure 3-6;

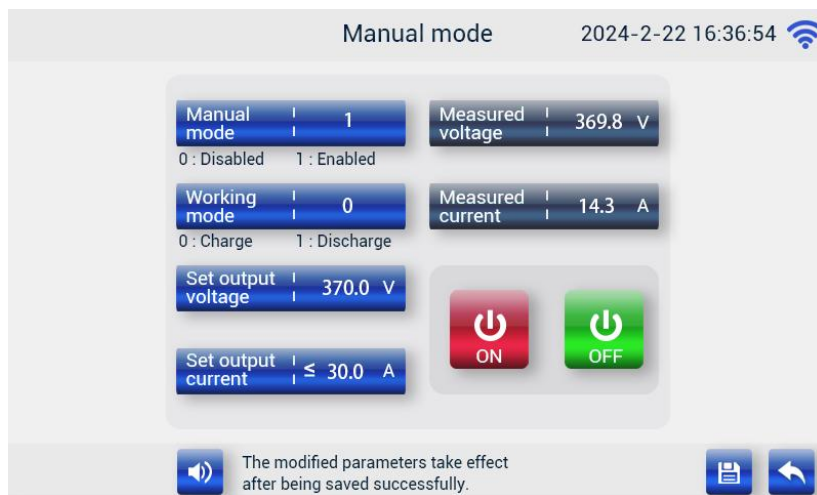


Figure 3-6 Manual Control Interface

- **Manual control mode enable:** 0 represents disable, 1 represents enable manual control module for testing; **This function is used for testing and should be disabled when not in use, otherwise it will affect normal charging;**
- **Charge/discharge mode:** 0 represents charging, 1 represents discharging, and users can choose the direction of energy flow according to their needs;
- **High and Low Voltage Modes:** The voltage level during module operation can be set, with 200~500V being the low voltage mode and 500~1000V being the high voltage mode;
- **Voltage setting:** Set the target voltage output by the module, which must not exceed the rated voltage of the module;
- **Maximum current limit:** Set the maximum current output of the module, which must not exceed the rated current of the module;
- **Current voltage:** The module's current output voltage can be viewed in real-time;
- **Current Current:** The module's current output current can be viewed in real-time;
- **Power on/off button:** ON represents opening the module, OFF represents closing the module; When modifying voltage and current parameters during testing, there is no need to stop, simply modify and save the parameters;

DC side port parameter settings



- **DC voltage:** displays the actual voltage value of the current DC side port;
- **DC current:** displays the actual current value of the current DC side port;
- **Busbar voltage:** displays the actual voltage value of the intermediate bus inside the current module;
- **DC minimum operating voltage setting value (Vbus Min voltage):** In charging mode, the lower limit of the allowed operating voltage on the DC side. When the actual DC voltage value is lower than this value, the charging power will be reduced or even stopped to prevent battery/PCS overdischarge;

- **DC maximum operating voltage setting value (Vbus Max voltage):** In discharge mode, the upper limit of the allowed operating voltage on the DC side. When the actual DC voltage value is higher than this value, the discharge power will be reduced and switched to constant voltage discharge to prevent overcharging of the battery/PCS;
- **Power derating value:** indicates the maximum output power output capacity of the current system. Under normal circumstances, it will display full power (such as 22Kw). If there is an over temperature derating or DC side voltage undervoltage derating (refer to the DC side power curve), this value will be adjusted and reduced according to the actual situation;
- **DC Maximum Charge Current:** In charging mode, it indicates the maximum discharge current of the DC side battery/PCS;
- **DC maximum discharge current:** In discharge mode, it indicates the maximum charging current of the DC side battery/PCS;
- **PCS Type:** 0 represents the connection to the low-voltage PCS system (maximum operating voltage of 570V on the DC side), and 1 represents the connection to the high-voltage PCS system (maximum operating voltage of 980V on the DC side); This function is used to distinguish the systems connected to the DC side, and users can set the corresponding PCS type according to their actual situation.

Network Settings

Click the network settings button to enter the network related settings page, as shown in Figure 3-7;

The figure shows three screenshots of the network settings interface. The top screenshot shows the IP settings page with fields for Internet (1), Local IP (192.168.1.130), Subnet (255.255.255.0), Gateway (192.168.1.1), Remote IP (192.168.1.10), Port No. (2000), MAC add. (0-128-255-0-0-0), and Serial No. (EMG01DC120). The middle screenshot shows the Security settings page with fields for Remote URL (0), URL setting (coreapp.elmotion.ro/websocket/CentralSystemService), FTP URL (ftp://192.168.1.189), and Firmware Upgrade (0). The bottom screenshot shows the Modbus TCP Slave settings page with fields for License (0000020101501010000000000000122), Identification (aleID7FEI9), V2G name (015050201011000000000000224), and Secret KEY (2a577c5acc4e7b3b15a53d3017f32bd9).

Figure 3-7 Network parameter settings

① IP settings

- **Communication method:** This monitoring system can currently achieve Ethernet communication (ETH), WiFi communication, and 4G communication; If network communication input 0 is not required; If you choose WiFi communication, you need to

click the green WiFi button to enter the WiFi selection interface, as shown in Figure 4-8. The software will automatically search for WiFi and refresh every 15 seconds, or manually click the refresh button; Select the WiFi to connect to and query, and enter the correct password (8 characters or more) in the pop-up window to connect. The displayed password will be encrypted; If the WiFi is successfully connected, the password will be automatically saved for the next connection. You can view or delete the saved WiFi password in "Advanced Settings", up to 10 passwords can be saved; If WiFi is hidden, you can manually "add network" and enter the WiFi SSID and password to connect, as shown in the figure;



Figure 3-8 WiFi Connection Settings

- **Local IP address, subnet mask, and default gateway:** Customers do not need to manually set them, and the monitoring board dynamically obtains the IP address;
- **Remote IP address:** If the URL setting has enabled activation, the remote IP address will be automatically obtained from the server; If it is not activated, it needs to be manually set. This address is the IP address of the target server, such as 120.77.82.99;
- **Port number:** is the port number of the target server, such as 2000;
- **MAC address:** It is the physical address of the Ethernet monitoring board. If using a local area network, it can be randomly set here, but it is best not to conflict between monitoring boards. This address is unique, and the monitoring will also send random values. This does not need to be set, and the MAC address displayed on the screen is in decimal;
- **Chargerserial number setting:** Each V2X charger has a unique Chargerserial number that cannot be repeated. After setting, restart the V2X charger;

② URL settings

- **Activate URL setting:** When set to 1, the Charger will automatically obtain the remote IP address obtained from the server and display it on the pile. If set to 0, the remote IP address needs to be manually set;
- **URL setting:** This feature enables automatic networking after changing the domain name address; Enter the corresponding domain name address in the URL settings box, enable this function, click save, enter IP settings, modify the corresponding port number and Charger number, save the parameters, exit, and the remote IP will be automatically updated and connected to the network;
- **FTP domain name setting:** This feature can be used to upgrade firmware by building an FTP connection; Enter the FTP domain name, select the firmware code that needs to be upgraded, and click 'Upgrade' to complete the process; It is emphasized here that DC and PFC are power module software, and upgrading requires some time. After clicking the upgrade button, you need to exit to the homepage to achieve the upgrade. This upgrade method is generally not recommended to be used;
- **Firmware upgrade:** can be used for FTP firmware upgrade;

③ Device Information Settings

This feature is only available on the State Grid platform: Electric piles need to be registered before they can officially connect to the State Grid platform. There are two ways to register. The prerequisite is to first set the Charger number and communication method, and click save;

● **Method 1: Enter the registration code**

This method is suitable for first-time access to the State Grid platform. The registration code is provided by the State Grid platform, and the triplet (product identification, device name, device key) needs to be cleared on the screen before entering the registration code (when the triplet is not empty, it will be directly verified using the triplet). After the platform verification is passed, a triplet will be allocated to the power station, and then it will be connected to the platform to complete registration. The electric Charger will save the new triplet and display it on the screen, and the next connection will be directly connected to the platform using the triplet.

● **Method 2: Enter a triplet**

Triple: product identification, device name, device key, directly enter the triple and connect to the platform to complete registration.

Both methods ultimately complete registration through triplets, and once verified, the platform can be accessed.

④ **Security settings**

This function is only used for OCTT testing;

⑤ **Modbus TCP settings**

IP address: Modbus TCP local address

GateWay: Gateway address

IP Mode: How to obtain an IP address, 0. Static IP address; 1. Dynamic IP address;

Port No: Modbus TCP Port Number Setting

Note: After setting the network parameters, it is necessary to restart the V2X charging and discharging station to ensure that all parameters of the monitoring board comply with the designed V2X charging and discharging station scheme;

Module operating condition interface

Click the module operating condition button to view the working condition of each charging module, as shown in Figure 3-9; The monitor performs polling operations on each module through the CAN bus to obtain the current number of online modules, as well as the current output voltage, current, ambient temperature, module online status, and other information of each module. Currently, it can display information for up to 20 modules;

PMs condition					
PM Address	Ouput voltage(V)	Ouput current(A)	Working TEMP (°C)	Communication	PM status
1	369.0	22.2	22.0		
2	0.0	0.0	0.0		
3	0.0	0.0	0.0		
4	0.0	0.0	0.0		
5	0.0	0.0	0.0		
6	0.0	0.0	0.0		
7	0.0	0.0	0.0		
8	0.0	0.0	0.0		
9	0.0	0.0	0.0		
10	0.0	0.0	0.0		

1
online

1/2

Figure 3-9 Module operating condition information interface

Factory testing interface

Click the factory test button to test the DC contactor, card reader, electronic lock, and other conditions. The interface is shown in Figure 3-10; The testing content on this page can determine whether the corresponding components and monitoring units on the system are intact.



Figure 3-10 Factory Test Interface

Record Inquiry

Record queries are divided into alarm record queries and charging record queries. The alarm record interface is shown in Figure 3-11, and the charging record interface is shown in Figure 3-12; The alarm record mainly records some faults that occur during standby or charging, and will form a record in a timely manner, saving up to 90 latest alarm records; Each time the charging is completed, a charging record will be saved, mainly recording the start time of charging, the card number used for charging, and the reason for ending the charging. The latest 100 charging records can be saved at most;

Alarm record			2023-4-3 16:36:54	Charging record				2023-4-3 16:36:54
No.	Time	Alarm contents		No.	Time	Card No.	Record contents	
1	2023-4-3 16:32:33	Module communication failed!		1	2023-4-3 16:32:33	123456787654321	GunA swipe card to stop charged!	🔍
2	2023-4-3 15:24:31	Module communication recover!		2	2023-4-3 15:24:31	123456787654321	GunA swipe card to stop charged!	🔍
3	2023-4-3 15:12:43	Module communication is abnormal.		3	2023-4-3 15:12:43	123456787654321	GunA swipe card to stop charged!	🔍
4	2023-4-3 15:09:54	Electroic lock status recover!		4	2023-4-3 15:09:54	123456787654321	BMS identification failed.	🔍
5	2023-4-3 13:42:13	Electroic lock status is abnormal.		5	2023-4-3 13:42:13	123456787654321	GunA swipe card to stop charged!	🔍
6	2023-4-3 13:36:21	Module communication failed!		6	2023-4-3 13:36:21	123456787654321	AC input undervoltage.	🔍
7	2023-4-3 10:24:39	Module communication recover!		7	2023-4-3 10:24:39	123456787654321	GunA swipe card to stop charged!	🔍
8	2023-4-3 12:42:43	Module communication is abnormal.		8	2023-4-3 12:42:43	123456787654321	GunA user can stop charged!	🔍
9	2023-4-3 11:09:50	Electroic lock status recover!		9	2023-4-3 11:09:50	123456787654321	BMS identification failed.	🔍
10	2023-4-3 10:33:18	Electroic lock status is abnormal.		10	2023-4-3 10:33:18	123456787654321	GunA Setup Timeout!	🔍

Figure 3-11 Alarm Record

Figure 3-12 Charging Record

Click the information button on the right side of each charging record to enter the charging details interface, where users can view card number, charging duration, electricity consumption, cost, start and end SOC, charging mode and working mode, vehicle battery type, etc;

Detailed charging record

2025-8-6 14:02:55

Card NO. | 0000001234567890

Stop SOC | 90 %

Charging time | 35 min

Charging mode | CV

Charging power | 2.6 kWh

Battery model | LiFeP04

Fee amount | 3.6

Work mode | Charge

Start SOC | 20 %

Figure 3-13 Detailed Charging Record

4 Card swiping charging/discharging process

Charger setting

Set the corresponding parameters in Figure 3-5 according to the specifications and models of the purchased V2X charger. After setting and saving the parameters, restart the V2X charger to proceed with the card swiping charging/discharging process;

Start card swiping for charging/discharging

Firstly, ensure that the charging gun is connected. If the charging gun is not connected, it cannot enter charging/discharging mode. There will be a prompt on whether the car gun is connected to the homepage;Click SOC to set the SOC value at the end of discharge;If the charging gun is connected, you can click the "swipe card" button to jump to the swipe card interface, as shown in Figure 4-1;If the card swipe fails, an abnormal pop-up window will appear. Find the abnormality and swipe the card again;After swiping the card correctly, enter the V2X charger self-test interface.

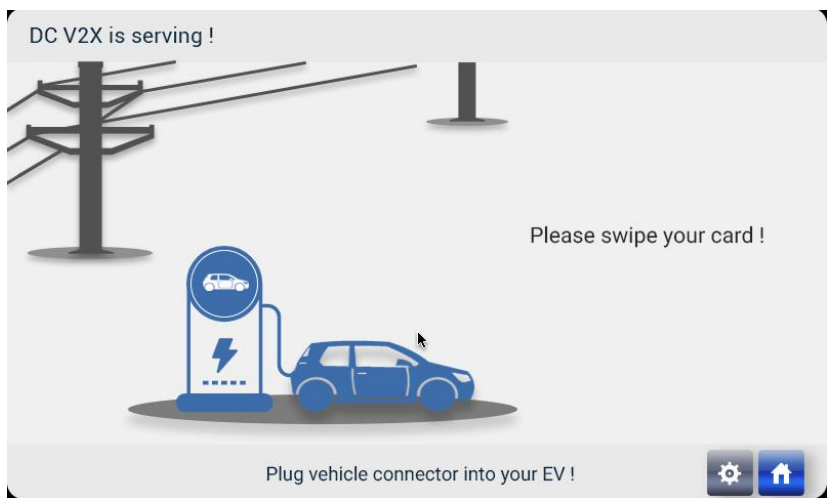


Figure 4-1 Card swiping charging/discharging page

System self-test in progress

After successfully swiping the card, enter the self-test interface, as shown in Figure 4-2. The self-test includes monitoring the insulation resistance value, contactors, input overvoltage and undervoltage, and communication status of the equipment.If the self-test is abnormal, the status indicator light will turn red or the display screen will show the corresponding fault information, while forming a fault record;

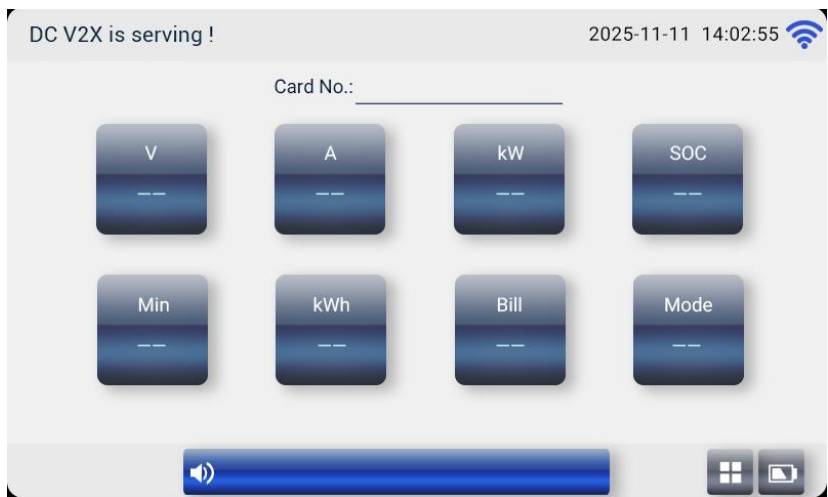


Figure 4-2 System self-test

Charging/discharging in progress

If the V2X charger self-test is normal, enter the charging/discharging interface, as shown in Figure 4-3. During the charging/discharging process, users can check the charging/discharging voltage, current, battery level, and time at any time, or query the battery information of the electric vehicle and the working status of the entire charging system power module. Click on "Battery Information" and "Module Operating Conditions" respectively to view the status information. The green light indicates normal, and the red light indicates abnormal. As shown in Figure 4-4:

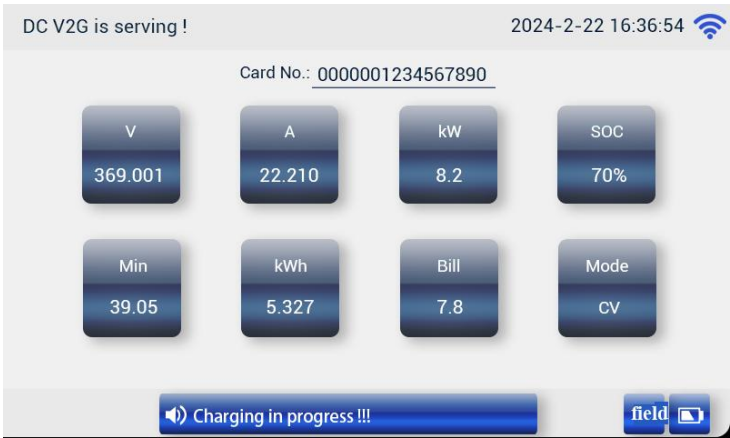


Figure 4-3 Charging/discharging in progress page



Figure 4-4 Battery Information Page (left) and Module Operating Interface (right)

Swipe card to end charging/discharging

If the charging/discharging process needs to be stopped, the user can simply swipe the card and end it. After swiping the card, the user will be redirected to the settlement interface, which displays detailed information such as card number, battery level, and time for confirmation; If the fault stops or the charging/discharging is completed, the program will automatically jump and stay on the settlement page. Customers only need to correctly swipe their card to settle the current charging/discharging, and the entire charging/discharging process can be completed. Then, they can jump to the homepage to start a new charging/discharging operation again; The interface is shown in Figure 4-5 below:

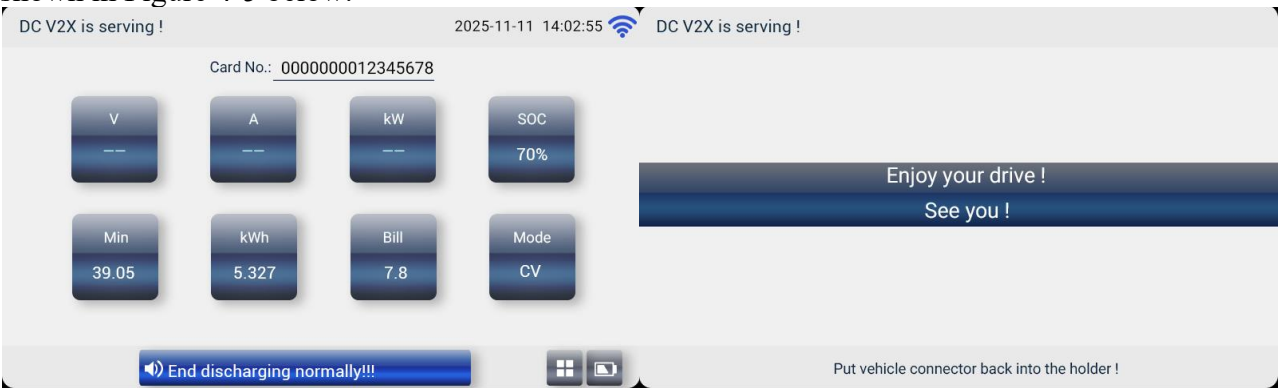


Figure 4-5 Card swiping ends charging

Figure 4-6 Goodbye interface

After swiping the card, the interface automatically jumps to goodbye, prompting the customer to pull out the gun and return it to its original position for easy charging/discharging next time. The interface is shown in Figure 4-6.

5 Alarm and fault handling methods

5.1 General fault handling process

The occurrence of alarms during installation and debugging is a normal phenomenon in this process. By mastering the general fault handling process, one can identify the root cause of the fault based on its symptoms, conduct analysis, and thus eliminate the fault.

5.2 Fault List and Solutions

alarm record	solution proposal
System lightning arrester malfunction	1. Check if the lightning protection feedback line is connected incorrectly; 2. Check if the lightning arrester is damaged;
Abnormal humidity	NA
Press the emergency stop button	1. Check if the emergency stop button has been pressed and restore the emergency stop button; 2. Check if the emergency stop signal is normal and try to restore it; 3. Check if the hardware of the emergency stop button is damaged. If it is damaged, replace the emergency stop button;
Abnormal communication of charging module	1. Check the parameter configuration and confirm the settings according to the module parameters; 2. Check if the module is powered normally, if the front-end power switch is closed, and use a multimeter to measure if there is voltage on the DC input side; 3. Check if the communication wiring between the monitoring board and the charging module is loose or reversed, and re plug and unplug it; 4. If the problem persists, please contact customer service and provide feedback to the module based on the fault displayed on the alert page that pops up;
Abnormal DC circuit breaker	NA
Abnormal DC surge protector	NA
Communication abnormality of A gun DC energy meter	1. Check if the DC energy meter type and protocol type set for parameter settings are correct; 2. Check if the communication wiring between the monitoring board and the DC energy meter is loose or reversed, and re plug and unplug it; 3. Check if the running light of the DC energy meter is flashing. If it is not flashing, check if the power supply wiring of the DC energy meter is loose and if it returns to normal after re plugging and unplugging; 4. If the problem still exists, can replacing the same DC meter restore normal operation;
Grid abnormality	1. Module failure, common causes: the module is powered on as soon as it is powered off; Note: When the module is powered off, it needs to wait for 10 seconds before it can be powered on;
The positive terminal status of the A-gun contactor is abnormal	1. In non charging state, use a multimeter to check the status of the DC contactor and determine if it is closed; 2. Check whether the DC contactor control in the parameter settings is normal, and whether the DC contactor can be controlled normally in the factory settings interface. If it cannot be controlled, the contactor is likely to be stuck, please replace it in a timely manner;
Abnormal status of door	1. Check if the parameter settings enable the door control switch and

control switch	if the Chargerdoor is not closed properly;
Communication abnormality of leakage current sensor	1. Check if the insulation communication wiring on the monitoring board is loose or reversed, and re plug and unplug it;
Communication abnormality of card reader	1. Check if the Charger does not have a card reader, and if the card reader is enabled in the parameter settings card type; 2. If there is a card reader, check whether the communication wiring between the monitoring board and the card reader is loose or reversed; 3. Check if the red running light of the card reader is always on. If it is not on, replace the card reader to see if it can restore normal operation;
DC input undervoltage	1. Check if the DC undervoltage alarm value on the parameter setting interface is reasonable; 2. Measure the DC incoming voltage with a multimeter. If the voltage is lower than the set value, immediately cut off the system power supply;
DC input overvoltage	1. Check if the DC overvoltage alarm value on the parameter setting interface is reasonable; 2. Measure the DC incoming voltage with a multimeter. If the voltage is higher than the set value, immediately cut off the system power supply;
A gun output undervoltage	1. Check whether the DC undervoltage alarm value setting on the parameter setting interface is reasonable; 2. Restart charging and check the output voltage on the vehicle or V2X charging and discharging station during the charging process. If the voltage is lower than the set value, the charging is unsuccessful;In this case, please contact customer service for a solution and do not continue charging;
A gun output overvoltage	1. Check whether the DC overvoltage alarm value setting on the parameter setting interface is reasonable; 2. Restart charging and check the output voltage on the vehicle or V2X charging and discharging station during the charging process. If the voltage is higher than the set value, the charging is unsuccessful;In this case, please contact customer service for a solution and do not continue charging;
A gun output overcurrent	1. Check if the DC overcurrent alarm value setting on the parameter setting interface is reasonable, 2. Check whether the current value displayed by the module and the current value detected by the meter are normal;
A gun insulation detection fault	1. The insulation resistance value of the system DC+to ground or DC - to ground is too low. Check whether the charging gun wire is aging and whether the gun head is flooded; 2. Check if the insulation resistance value at the vehicle end is normal;
Abnormal insulation detection of gun A	1. The insulation resistance value of the system DC+to ground or DC - to ground is low, but it does not affect current use.It is necessary to check whether the charging gun cable is aging and whether the gun head has water ingress; 2. Check if the insulation resistance value at the vehicle end is normal;
Group A module type mismatch	NA
Module address setting error	NA
System fan malfunction	NA
Press the S3 switch of gun A	NA
Abnormal fusion terminal	NA

communication	
The positive terminal status of the A-gun contactor is abnormal	1. In non charging state, use a multimeter to check the status of the DC contactor and determine if it is closed; 2. Check whether the DC contactor control in the parameter settings is normal, and whether the DC contactor can be controlled normally in the factory settings interface. If it cannot be controlled, the contactor is likely to be stuck, please replace it in a timely manner;
The negative terminal status of the A-gun contactor is abnormal	1. In non charging state, use a multimeter to check the status of the DC contactor and determine if it is closed; 2. Check whether the DC contactor control in the parameter settings is normal, and whether the DC contactor can be controlled normally in the factory settings interface. If it cannot be controlled, the contactor is likely to be stuck, please replace it in a timely manner;
A gun CPU fan status is abnormal	1. Check if the CPU fan wiring is normal, and after re plugging and unplugging, observe if the fan status is abnormal;
GunA CP signal short circuit!!!	1. After inserting the gun, check if the CP voltage is at the normal value of 9V. If it is 0V, it is a CP short circuit and the CP terminal circuit needs to be checked for normal operation;

5.3 System fault handling

5.3.1 protect

- DC input overvoltage, undervoltage, phase loss, and overheating will activate protection, which can be confirmed by the fault code.
- The cabinet may be protected against overheating due to lack of ventilation or high temperature in the computer room environment.

5.3.2 fault

- High output voltage or overcurrent will cause faults. In this case, disconnecting the DC power and restarting should restore normal operation.
- When in automatic working mode, the output voltage is controlled by the upper computer. If there is monitoring at this time, the output voltage may not be the same as the actual set value, which is caused by the temperature compensation function and is a normal phenomenon.

5.3.3 uneven flow

- Confirm that the output current is above half load, otherwise it is meaningless.
- Failure to connect the equalizing line or incorrect connection of the equalizing line may result in uneven current.
- Disconnect the current distribution line and communication line, load, and measure the signal on the current distribution port. The magnitude of the signal should meet the requirement of $i/1.05I \times 4.2V$, where i is the actual output current and I is the rated output current; At this point, short-circuit the positive and negative terminals of the current sharing port, and the output voltage should drop by about 10V.
- To check the voltage and current switch in CAN communication/current sharing, an additional 120 ohm resistor is required. Press the voltage and current switch button to see if the voltage and current are displayed synchronously.

5.3.4 Communication interruption

- Incorrect address settings will result in communication interruption, and two different addresses with the same settings will also cause monitoring communication interruption.
- In non working state, it will cause communication interruption.
- When the number of monitors set exceeds the actual number, it will result in setting unnecessary communication failures because they do not exist at this time.
- The address requirement starts from 1 and is set continuously within the same group.

5.3.5The voltage output cannot reach the set voltage

- Overload will cause current limitation, preventing the output voltage from reaching the set value.
- In the float charging state, the temperature compensation function will cause the output voltage to be inconsistent with the set value.

5.3.6Report input exception

- If using a small bracket, check if it is plugged in properly and if the power supply is properly connected.
- If there is insurance, check if there is insurance coverage.
- If necessary, pull each wire of the power supply inlet by hand to see if there is any virtual connection.

5.3.7Unrestricted flow

- The battery current sampling wiring is incorrect (such as the battery current sampling line being connected to the control bus current terminal).

5.4 Troubleshooting of Monitoring Module

5.4.1LCD keeps flashing or has no display

- Check if the power input is normal (12V), if the polarity is correct, and if the wiring is correct.

5.4.2The interface display is not clear

- You need to adjust the brightness in the "System Settings - Other Settings" column.

5.4.3Touch interface unresponsive and black screen

- Check if there is a 12V voltage at the power supply end of the monitoring unit. If there is no voltage, check if the power line collection is loose. If everything is normal, the monitoring unit needs to be replaced.

5.4.4The main monitoring system reports communication failures in all or part of the monitoring unit modules

- Check if the main monitoring settings are consistent with the actual configuration.
- Check whether the connection wires of all RS485 interfaces are correct and the wiring is good.
- Disconnect all RS485 ports and check if each unit's RS485 port has a voltage of around 3V. If it is not normal, replace the corresponding unit.
- The main monitoring system communicates with each unit one by one. If it cannot communicate with any of the units, it needs to be replaced;If communication with all units is not possible, the main monitor needs to be replaced.

5.4.5The main monitoring and backend cannot communicate

- Check whether the communication protocol, address, and baud rate of the main monitoring are consistent with the background selection. If they are not consistent, they need to be changed to be consistent.
- Check if the RS485/RS232 mode selection is consistent with the background. Our company defaults to RS485/Modbus mode at the factory.
- Check whether the communication interface wiring is correct, whether it is reversed or short circuited. If it is incorrect, it needs to be changed.
- If some data is incorrect when communicating with the backend, please contact the backend manufacturer.
- If communication still cannot be achieved after following the above steps, the main monitor needs to be replaced.

5.5 DC input fault handling

- DC input circuit breaker tripping (if equipped with matching box and DC circuit breaker):
A short circuit in the DC input section may cause the DC input circuit breaker to trip.

5.6 DC output fault handling

- Total output trip:
Output overload may cause the main output switch to trip.
Choosing the wrong type of auxiliary contact may cause the above alarm.
The voltage of the output switch release is 12V. If it cannot be tripped, please check if the voltage is normal.
- Abnormal output voltage causing overvoltage or undervoltage alarm:
Voltage meter malfunction will cause abnormal bus voltage detection, resulting in overvoltage or undervoltage alarms;
Inappropriate setting of overvoltage or undervoltage alarm points will result in abnormal bus voltage alarms.
Check if the output DC contactor is engaged, the coil voltage of the DC contactor is 12V.

6 Upgrade and Diagnosis

6.1 Local USB drive upgrade instructions

Firstly, prepare the USB drive, preferably using the one that comes with our shipment. If using your own USB drive, the file system should support the FAT32 format

Create a uuapp folder on the USB drive and place the uuevseApp software we provided into the uuapp folder, as shown in Figure 1-figure4

Subsequently, the USB flash drive is inserted into the interface marked with an APP on the side of the pile. Three long beeps and one short beep are heard, indicating that the burning is complete. The USB flash drive is removed, and the Charger is powered off and restarted. Then go to the information interface to confirm whether the software version is the version number given by the engineer. specific operation

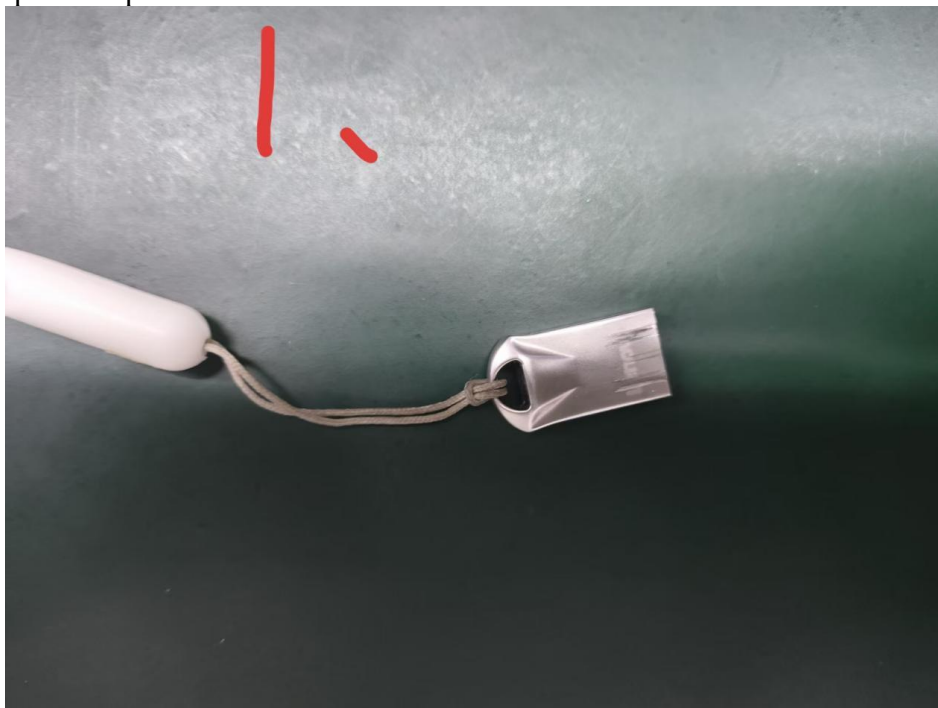


Figure1



Figure2



Figure3

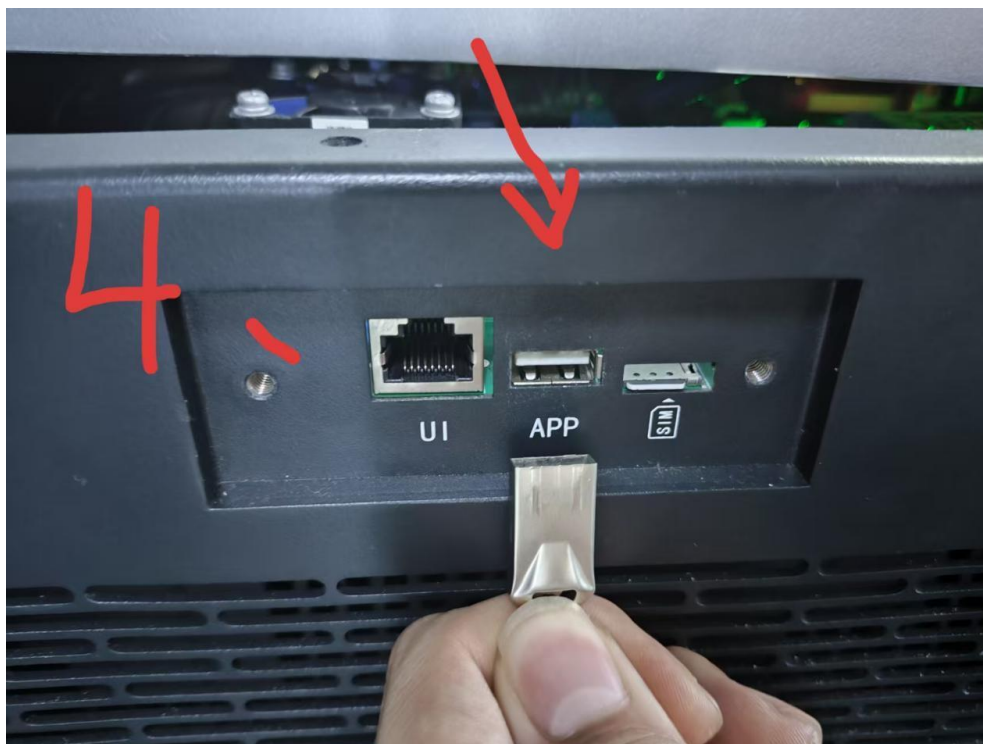
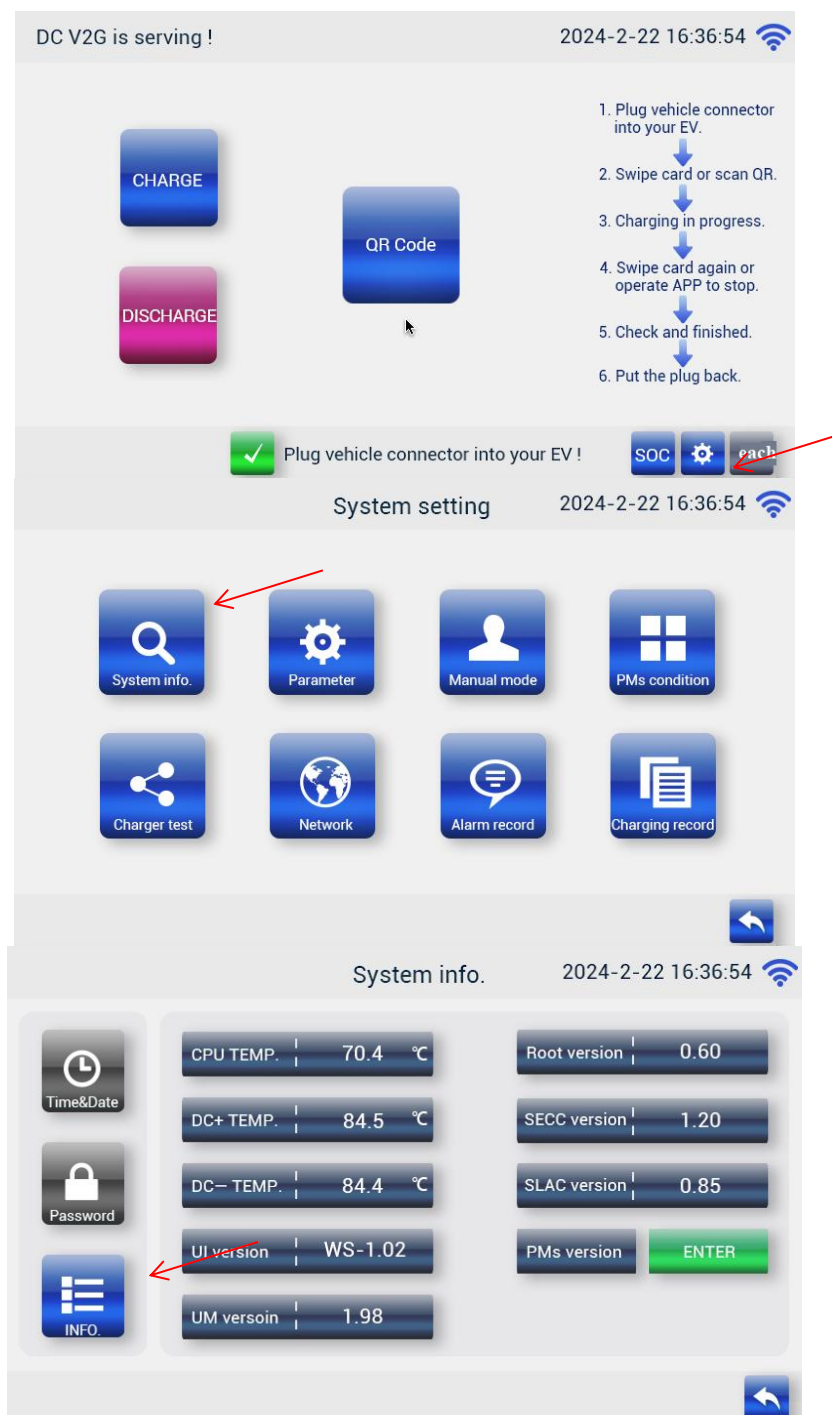


Figure4

The steps to confirm the software version are: Figure 5-Figure 8, where the default password is 123456



6.2 Remote OTA upgrade in the background

1. Supports FTP/FTP upgrades when connected to the internetThe inverter issues an upgrade command to trigger the OTA upgrade process;

2. In the networked state, it supports the upper computer software to upgrade and maintain the equipment.The networking method can be configured by the inverter, supporting wired ETH/WIFI. If WIFI networking is selected, the inverter can synchronize the router WIFI and password to the V2X charging station for networking;

3. The specific operation of OTA can be found in the attachment Modbus Protocol V2.3 4.1.7: Setting up Network and 4.1.8: FTP Upgrade.

7 Product packaging and transportation

7.1 packaging

The product is packaged in neutral cardboard or wooden boxes, lined with shock-absorbing pearl cotton and waterproof plastic bags.

7.2 transportation

The product should be covered during transportation and should not experience severe vibration or impact.

7.3 keep

The equipment should be stored in a well ventilated area, avoiding environments with high temperatures, dust, and metal powder. During storage and transportation, rain, snow, and severe vibration impacts should be avoided.

7.4 maintain

(1) Regular inspection

Check the dust accumulation of the anti heat sink, and if necessary, clean the heat sink to avoid affecting the heat dissipation effect; Check if the touch screen displays any faults and if the indicator lights are functioning properly;

Check whether the charging cable is damaged, whether the charging plug has foreign objects or water ingress, and whether the pole has oxidized or deformed; Check whether the internal of the charger is normal, whether there is water immersion, and whether the cable connection is loose;

Press the emergency stop button to check if the DC relay can be disconnected and reset after confirmation.

(2) Maintenance operation

The charging equipment is operated by professionals or trained personnel inside, and other personnel are prohibited from opening the charging equipment door; During maintenance, it is necessary to disconnect the higher-level DC switch and verify that there is no power at the input end to confirm safety before maintenance operations can be carried out; Press the emergency stop button in case of emergency, and only reset the emergency stop button after confirming and eliminating the fault;

It is strictly prohibited to replace internal components of the product without the consent of the manufacturer.

7.5 Contact Us

When encountering unresolved issues or requiring assistance, please contact us. Technical Support Hotline: Customer Service Department 18813909163

Business Contact Hotline: Marketing Department 18979110400

8 Attachment V2X Modbus Protocol V2.3

catalogue

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V2X-MODBUS communication protocol

1. Document description

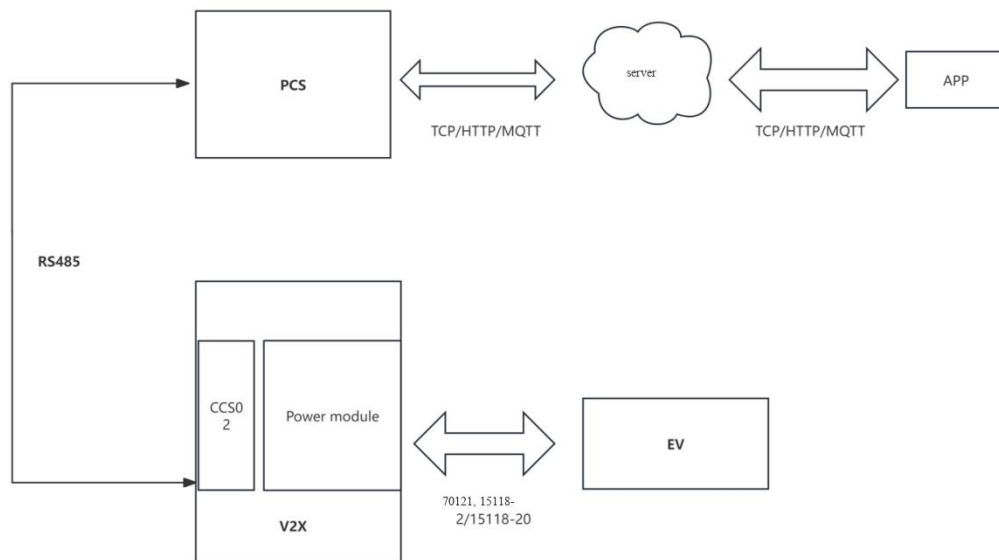
1.1 Purpose

Clarify the specifications for data exchange between devices through Modbus protocol, ensuring good compatibility and stability in communication between customer systems and devices.

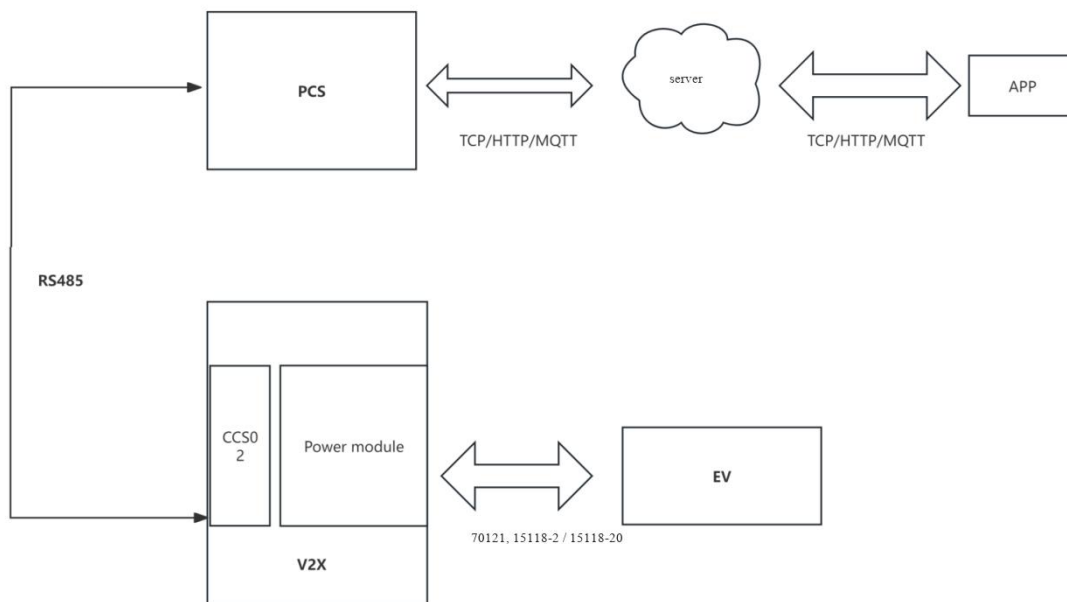
1.2 Scope of Application

Applicable to all third-party systems or devices that communicate with this device based on Modbus protocol, refer to the communication diagram below

If Modbus TCP is used, the physical layer communication method is usually wiredThe communication framework diagram is as follows:



If Modbus RTU is used, RS485 is usually used for communication at the physical layer, and the communication framework diagram is as follows:



2. Protocol Overview

2.1 Introduction to Modbus TCP

The Modbus protocol based on TCP/IP typically uses standard port 502 as its default port.

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Protocol features: good real-time performance, high reliability, and wide compatibility.

2.2 Introduction to Modbus RTU

Modbus RTU is an industrial protocol based on serial communication (such as RS-485/RS-32), using binary encoding, master-slave architecture, and supporting one master and multiple slaves communication

2.3 Role

If Modbus TCP is used, the role definition is as follows

Client: Send request (PCS/client system)

Server: Response request (V2X/local device)

If Modbus RTC is used, the character definition is as follows

Host: PCS/Customer System

Slave: V2X/local device

3. Agreement details

3.1 Data Frame Format

The Modbus TCP/RTU data frame structure is as follows:

field name	length	description	example
transaction identifier	2	Request/Response Matching Identifier	0x0001
Protocol identifier	2	Fixed as 0x0000	0x0000
length	2	Subsequent byte count (including unit ID)	0x0006
Unit ID	1	Device address (0x01/0x02)	0x01/0x02
Check code (CRC)	variable	Function code+data	0x03 0x00...

3.2 Function code definition

Code (hexadecimal)	function code	Operation Type	register type
0x03	Read and hold register	readable	holding register
0x04	Read input register	readable	input register
0x06	Write to a single register 注：绿能研发专用	writable	holding register
0x10	Write multiple registers	writable	holding register

4. Register Address Mapping Table

4.1 Keep Register

Supported operations: 0x03 is used to read multiple parameters, 0x06 is used to write a single parameter, and 0x10 is used to write multiple parameters.

4.1.1 Set device system time (address 0x01):

function code	register address	variable	R/W	unit	Data format	length	remark
	0x4001-0x4007						
0x03/ 0x06/ 0x10	0x4001	time zone	R/W	-12~ 12	Uint16	1	
	0x4002	year	R/W	0-99	Uint16	1	
	0x4003	moon	R/W	1-12	Uint16	1	
	0x4004	day	R/W	1-31	Uint16	1	
	0x4005	time	R/W	0-23	Uint16	1	

	0x4006	part	R/W	0-59	Uint16	1	
	0x4007	second	R/W	0-59	Uint16	1	

4.1.2 Set input/output alarm values (address 0x01):

function code	register address	variable	R/W	unit	Data format	length	remark
	0x4008 ~ 0x400B						
0x03/ 0x06/ 0x10	0x4008	Set output undervoltage value	R/W	V	Uint16	1	0~511
	0x4009	Set output overvoltage value	R/W	V	Uint16	1	0~4095
	0x400A	Set input undervoltage value	R/W	V	Uint16	1	0~511
	0x400B	Set input overvoltage value	R/W	V	Uint16	1	0~1023

4.1.2 Electricity price setting (address 0x01, **vehicle charging billing**):

function code	register address	variable	R/W	unit	Data format	length	remark
	0x400C ~ 0x4010						
0x03/ 0x06/ 0x10	0x400C	Wave electricity price	R/W	0.0001	Uint16	1	
	0x400D	Peak electricity price	R/W	0.0001	Uint16	1	
	0x400E	Flat electricity price	R/W	0.0001	Uint16	1	
	0x400F	Valley electricity price	R/W	0.0001	Uint16	1	
	0x4010	Electricity loss rate	R/W	%	Uint16	1	

4.1.3 Setting current limiting parameters/Setting discharge cut-off SOC value/Setting charge discharge mode (address 0x01):

function code	register address	variable	R/W	unit	data format	length	remark
	0x4011-4013						
0x03/ 0x06/ 0x10	0x4011	Set power limit	R/W	0-22000(W)	Uint16	1	Vehicle end parameter control
	0x4012	Set the discharge cut-off SOC (the lowest SOC value during discharge)	R/W	10~90(%)	Uint16	1	
	0x4013	Set charging and discharging modes (control mode)	R/W	0: Stop 1. charging Start 2. discharging Start	Uint16	1	

Switching between discharge modes is not currently supported during charging

4.1.4: Set to read charging record register (address 0x01, vehicle side historical charging and discharging records, 10000 records)

function code	register address	variable	R/W	unit	Data format	length	remark
	0X4014-4017						
0x03/ 0x06/ 0x10	0x4014	Total Count	R	-	Uint16	1	
	0x4015	Current record index	R/W	0-100	Uint16	1	0 does not display
	0x4016	Clear records	R/W	1: Clear	Uint16	1	

Current record index: XX charging records have been read;

Clear records: Clear all charging records, current record index: 0;After successful clearing, reset it to 0;

4.1.5: Set the Chargerparameter register (address 0x01)

function code	register address	variable	R/W	unit	Data format	length	remark
	0X4030-404F						
0x03/ 0x06/ 0x10	0x4030	Charging Chargercontrol	R/W	-	Uint16	1	
	0x4031~0x304F	retain	R/W				
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4.1.6: Set PCS information (address 0x01, input bus voltage parameter for EV pile, used for synchronous alarm and protection of EV pile)

function code	register address	variable	R/W	unit	Data format	length	remark
	0X4050-0X4080						
0x03/ 0x06/ 0x10	0x4050	Upper limit of VBus voltage setpoint	R/W	0.1V	Uint16	1	
	0x4051	Lower limit of Vbus voltage setpoint	R/W	0.1V	Uint16	1	
	0X4052	PCS system status	R/W	NA	Uint16	1	Receive operational status information from the photovoltaic inverter: 0. Shutdown; 1. Operation; 2. faults; 3. Standby; 4. Other

4.1.7: Set up network (address 0x01)

function code	register address	variable	R/W	unit	Data format	length	remark
---------------	------------------	----------	-----	------	-------------	--------	--------

	0X40C0-0X40FF						
0x03/ 0x06/ 0x10	0x40C0	Network mode	R/W	0: None 1: Ethernet 2: WIFI	Uint16	1	
	0x40C1	port number	R/W	0.1V	Uint16	1	
	0x40C2	Server IP address	R/W	NA	Uint16	2	For example, C0 A8 01 05 indicates that the IP is 192.168.1.5;
	0x40C4	WIFI account length	R/W		Uint16	1	Account character length, maximum 36
	0x40C5	WIFI account	R/W		Int16	18	
	0x40D7	WIFI password length	R/W		Uint16	1	Password character length, maximum 32
	0x40D8	WiFi password	R/W		Int16	16	password
	0x40E8	WIFI connection	R/W	Default 0	Uint16	1	Connect using the WIFI account password in the register, and it will automatically return to 0 regardless of whether the connection is successful or failed
	0x40E9	reserve	R/W		Uint16	3	
	0x40EC	充电器序列号 Charger serial number	R/W		Uint16	8	
	0x40F4-40FF	reserve	R/W		Uint16	12	

4.1.8: FTP Upgrade (Address 0x01)

function code	register address	variable	R/W	unit	Data format	length	remark
	0X4100-0x414F						
0x03/ 0x06/ 0x10	0x4100	URL	R/W		int16	64	
	0x4140-0x414C	reserve	R/W		Uint16	13	
	0x414D	upgrade type	R/W	0: None 1: APP 2:SECC 3:SLAC 5:DC.bin 6:PFC.bin	Uint16	1	
	0x414E	URL character length	R/W		Uint16	1	Maximum 128
	0x414F	Start upgrading/uploading	R/W	1: Start upgrading 3: Start uploading logs	Uint16	1	Starting the upgrade after setting it to 1, whether the upgrade is successful or failed, it will automatically reset to 0

4.2 Input Register

Supported operations: 0x04

4.2.1. Equipment information (address 0x01, **real-time running status** of EV pile):

Function al n-code	register address	variable	R/W	unit	data format	length	attention
	0x3001-0x301F						
0x04	0x3001	Software version (firmware version)	R	-	Uint16	1	
	0x3002	Module EV side information	R	-	Uint16	1	
	0x3003	reserve	R	-	Uint16	1	
	0x3004~0x300B	SN code (serial number)	R	-	Uint16	8	
	0x300C	CPU temperature (internal temperature of charger)	R	0. 1°C	Uint16	1	
	0x300D	Equipment status (status of EV charging equipment) 优优绿能研发专用	R	0. Standby mode 1. Charging in progress 2. During discharge 3. System malfunction 4. Charging/discharging completed 5. Locking	Uint16	1	
	0x300E	Connector status (EV Connector status)	R	0=No gun inserted, 1=Inserting a gun	Uint16	1	
	0x300F	reserve					
	0x3010	FTP upgrade status	R	0: No upgrade action 1: Upgrade successful 2: Upgrade request failed, please check URL 3: Device in operation, awaiting execution 4: Upgrading in progress 5: Download failed 6: Upgrade failed 7: The URL length is incorrect 8: Not connected to the internet	Uint16	1	
	0x3011	Network connection status	R	0: No network connection command 1: Connected successfully 2: Not connected to the platform, but	Uint16	1	

				networked 3: Not connected to the platform, not networked 4: Unable to find WiFi			
	0x3012	Network communication method	R	0: None 1: Ethernet 2: WIFI 3: 4G	Uint16	1	
	0x3013	local IP address	R		Uint16	2	0xC0A8 0x01B8 The representative IP address is 192.168.1.184
	0x3015~0x301F	retain					

4.2.2 Module Information (Address 0x01, Real time Operation Data Information of EV Pile):

function code	register address	variable	R/W	unit	data format	length	attention
	0x3020 ~ 0x303F						
0x04	0x3020	EV side voltage	R	V	Uint16	1	
	0x3021	EV side current	R	A		1	
	0x3022	reserve	R	V		1	
	0x3023	power factor	R	-		1	
	0x3024	Module Status	R			1	0. Standard 1. Other things
	0x3025	Module Temperature	R	0. 1°C		1	
	0x3026	Maximum output voltage	R	V		1	
	0x3027	Maximum output current	R	A		1	
	0x3028	rated power	R	W		1	
	0x3029	PM fault information 1	R			2	
	0x302B	PM Fault Information 2	R			2	
	0x302D	PM Fault Information 3	R			2	
	0x302F	DC side voltage	R	0.1V		1	
	0x3030~0x303F	retain					

Note: The charging details and vehicle side details are only valid during the charging/discharging period. In other states, the register value is 0.

4.2.3 Charging Details (Address 0x01, Real time Data Details at the Vehicle End)

function code	register address	variable	R/W	unit	data format	length	remark
	0x3040 ~ 304F						
0x04	0x3040	Charge discharge mode	R	0 None 1. Charging 2 Discharge	Uint16	1	
	0x3041	Charging voltage (Charger output voltage)	R	V	Uint16	1	
	0x3042	Charging current (Charger output current)	R	A	Uint16	1	
	0x3043	Power (active power)	R	W	Uint16	1	
	0x3044	Vehicle end SOC (State of Charge (SoC) EV)	R	%	Uint16	1	
	0x3045	Battery Level - Charged	R	0.001KWH	Uint 32	2	
	0X3047	Electricity level - discharged	R	0.001KWH	Uint 32	2	
	0x3049	Charging time	R	part	Uint16	1	
	0x304A	discharge time	R	part	Uint16	1	
	0x304B	优优绿能研发专用					
	0x304C	Startup status	R	0. NA 1. handshake 2. Parameter Configuration 3. self-check 4. precharge 5. Charging/dis charging in progress 6. stop	Uint16	1	
	0x304D~0x304F	retain					

4.2.4 Vehicle End Details (Address 0x01, Vehicle End Data Details):

function code	register address	variable	R/W	unit	data format	length	remark
	0x3050 - 0x306F						
0x04	0x3050	Charging protocol type	R	0 DIN70121 1 ISO15118-2 2 others	Uint16	1	
	0x3051	Remaining charging time	R	minute	Uint16	1	
	0x3052	Required voltage	R	V	Uint16	1	
	0x3053	Current demand	R	A	Uint1 6	1	
	0x3054	Maximum allowable charging point voltage	R	V	Uint1 6	1	
	0x3055	Maximum allowable charging current	R	A	Uint1 6	1	

0x3056	The maximum temperature of a single battery cell	R	0. 1°C	Uint16	1	
0x3057	Minimum temperature of a single battery	R	0. 1°C	Uint16	1	
0x3058	Maximum voltage of a single battery	R	V	Uint16	1	
0x3059	Minimum voltage of a single battery cell	R	V	Uint16	1	
0x305A	DC+Temperature	R	0. 1°C	Uint16	1	
0x305B	DC Temperature	R	0. 1°C	Uint16	1	
0x305C ~ 0x306F	retain					

4.2.5 System Fault Information (Address 0x01, **Fault Details of EV Pile**):

function code	register address	variable	R/W	unit	Number format	length	remark
0x3070 - 0x308F							
0x04	0x3070	PM status (fault code)	R	0. Normal 1. Fault	Uint16	1	
	0x3071	Fault status 1 LLC side	R		Uint32	2	
	0x3073	Fault state 2 BuckBoost side	R		Uint32	2	
	0x3075	Fault status 3 BuckBoost side	R		Uint32	2	

4.2.5 Charging Record Information (Address 0x01, **Detailed Charging Record at the Vehicle End**):

function code	register address	variable	R/W	unit	data format	length	remark
0x3090 - 0x30AF;							
0x04	0x3090	User card ID	R		Uint8	4	
	0x3094	SECC startup	R	1. Charging, 2. discharge	Uint16	1	
	0x3095	Charging capacity	R	0.001KW	Uint16	2	
	0x3097	Charging time	R	part	Uint16	1	
	0x3098	Discharge capacity	R	0.001KW	Uint16	1	
	0x309A	discharge time	R	part	Uint16	1	
	0x309B	Start SOC	R	%	Uint16	1	
	0x309C	Stop SOC	R	%	Uint16	1	

The non-existent charging record with UserCardID 0;

The ASCII code for charging and discharging without a card is 00000000, and the corresponding registers are 0x30 and 0x30

5. communication parameters

5.1 MODBUS-TCP parameters

The IP address of the IP/port device is xxx.xxx.xxx.xxx (visible on the screen), and the default setting is

The port number is 502.

Transmission Protocol: TCP.

ConnectTimeout: Exceeding 60 seconds;

5.2 MODBUS-RTU Parameters

The baud rate is 9600

6. Data format and encoding

Byte order: Big endian mode.

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Floating point encoding: Following the IEEE754 standard (e.g. 0x43, 0x7A, 0x00, 0x00) to represent floating point numbers 250.0.

7. Testing steps

Please use the ping command to confirm the network connection and ensure that the IP addresses of the main station and the device can communicate with each other. For example, enter ping [destination IP address] in the command prompt, and then press enter. If a reply is received, it indicates that the network connection is normal; If a request timeout occurs, there may be a network issue.

Configure from device Set the device address (unit ID) to 1.
information

Execute code: Call 0x03 (read) or 0x10 (write) as needed.

Test and verify the use of tools such as Modbus Poll to send requests, and check the response data.

8. Example requests and responses

Example 1: Read and hold register (function code 0x03)

Request: 35 47 00 00 00 06 01 03 40 01 00 02
Reply: 35 47 00 00 00 07 01 03 04 00
00 00 05

Example 2: Write to hold register (0x06)

Request: 00 11 00 00 00 06 01 06 40
01 00 02
Reply: 00 11 00 00 00 06 01 06
40 01 00 02

Example 3: Read input register (function code 0x04)

Request: 00 74 00 00 00 06 01 04 30 01 00 01
Response: 00 74 00 00 00 05 01 04 02 01 20

9. Error Handling

Exception response code:

error code	description
0x01	Illegal function code
0x02	Illegal data address
0x03	The data value exceeds the range

Example exception request: 00 C3 00 00 00 06 01 30 01 00 01

Example response status: 00 C3 00 00 00 03 01 81 01

10. Testing and Confirmation (to be supplemented)

Recommended testing tools:

Modbus polling (master station simulation)

Appendix A: Fault Codes (Overview of Fault Codes)

fault state variable	bit number	name	Fault Level (LCD screen attention, the upper computer does not require attention	remark
Fault status 1 LLC side	0	C1 wave by wave current limiting	Global Fault	
	1	C2 wave by wave current limiting	Global Fault	
	2	battery short circuit	Global Fault	
	3	Rapid overvoltage of battery	Global Fault	
	4	LLC1 fast overcurrent	Global Fault	
	5	LLC2 fast overcurrent	Global Fault	
	6	Rapid overcurrent of battery	Global Fault	
	7	reserve	N.A	
	8	Auxiliary source fault	Global Fault	
	9	Battery connection failure	Global Fault	
	10	battery-free	Global Fault	
	11	Battery overvoltage	Global Fault	
	12	Low battery voltage	Global Fault	
	13	Battery overcurrent	Global Fault	
	14	Uneven voltage of battery	Global Fault	
	15	reserve	N.A	
	16	电池未能成功建立母线 battery failed to establish busbar	Global Fault	
	17	LLC abnormality	Global Fault	
	18	Port Vbat input exception	Global Fault	
	19	CV setting exception	Global Fault	
	20	LLC current imbalance	Global Fault	
	21	reserve	N.A	
	22	C side radiator inlet overheating	Global Fault	
	23	C side radiator outlet overheating	Global Fault	
	24	DC side AD signal sampling abnormality	Global Fault	
	25	Battery relay malfunction	Global Fault	
	26	DC side E2PROM fault	Global Fault	
	27	485 communication failure	Global Fault	
	28	SPI communication failure	Global Fault	
	29	AN communication malfunction	Global Fault	
	30	DC side program burning error	Global Fault	
	31	Program version mismatch	Global Fault	
Fault state 2 BuckBoost side	0	Port battery short circuit	Global Fault	
	1	Bus voltage short circuit	Global Fault	
	2	BuckBoost wave by wave current limiting	Global Fault	
	3	BuckBoost abnormal wave generation	Global Fault	
	4	reserve	N.A	

	5	reserve	N.A	
	6	pid overvoltage of module busbar	Global Fault	
	7	Half module busbar rapid overvoltage	Global Fault	
	8	Port battery rapid overvoltage	Global Fault	
	9	Half port battery experiences rapid overvoltage	Global Fault	
	10	pid overcurrent of inductor current	Global Fault	
	11	reserve	N.A	
	12	reserve	N.A	
	13	Soft start relay malfunction	Global Fault	
	14	Main relay malfunction	Global Fault	
	15	reserve	N.A	
	16	Bus overvoltage	Global Fault	
	17	Port battery overvoltage	Global Fault	
	18	No battery malfunction	Global Fault	
	19	reserve	N.A	
	20	reserve	N.A	
	21	Unbalanced module bus voltage	Global Fault	
	22	Port battery voltage imbalance	Global Fault	
	23	Inductance current 1 segment overcurrent	Global Fault	
	24	Inductance current 2-segment overcurrent	Global Fault	
	25	Port battery current 1 segment overcurrent	Global Fault	
	26	Port battery current 2-segment overcurrent	Global Fault	
	27	Unbalanced inductor current	Global Fault	
	28	Module bus undervoltage	Global Fault	
	29	Port battery segment 1 undervoltage	Global Fault	
	30	Port battery 2-segment undervoltage	Global Fault	
	31	reserve	N.A	
Fault status 3 BuckBoost side	0	Port battery failed to establish bus	Global Fault	
	1	Port reverse connection fault	Global Fault	
	2	reserve	N.A	
	3	reserve	N.A	
	4	reserve	N.A	
	5	Buckboost side level 1 overload	Global Fault	
	6	Buckboost side 2nd level overload	Global Fault	
	7	PCS connection abnormality	Global Fault	
	8	Internal environment overheating	Global Fault	
	9	External environment overheating	Global Fault	
	10	buckboost Side radiator inlet overheating	Global Fault	
	11	buckboost Side radiator vent overheating	Global Fault	

	12	reserve	N.A	
	13	reserve	N.A	
	14	reserve	N.A	
	15	reserve	N.A	
	16	Abnormal sampling of module bus voltage	Global Fault	
	17	Abnormal sampling of port battery voltage	Global Fault	
	18	Abnormal port current sampling	Global Fault	
	19	Abnormal sampling of inductor current	Global Fault	
	20	Internal fan malfunction	Global Fault	
	21	External fan malfunction	Global Fault	
	22	MC side 485 communication fault	Global Fault	
	23	PCS communication failure	Global Fault	
	24	PCS malfunction itself	Global Fault	
	25	Buckboost side E2PROM fault	Global Fault	
	26	Work mode setting error	Global Fault	
	27	Buckboost side program burning error	Global Fault	
	28	reserve	N.A	
	29	reserve	N.A	
	30	reserve	N.A	
	31	reserve	N.A	

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